Queensland Guidance Note QGN36

Managing welding fume in coal mines

Date	December 2025
Revision	**1.0**

Resources Safety & Health Queensland

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1.0 Purpose

This guidance note provides practical advice for coal mine operators, supervisors and coal mine workers (CMWs) on managing exposure to welding fumes from hot work processes in Queensland coal mines. It aims to provide relevant, practical information specific to Queensland coal mines, without duplicating existing technical guidance.

Information is provided on:

- exposure profiles of CMWs involved in hot work
- common hot work processes and consumables in the coal industry
- risk management principles for identifying, assessing, and controlling exposure
- monitoring methods for evaluating personal exposure risk
- practical strategies for controlling fume
- health surveillance
- educational requirements for CMWs on health effects and control measures
- review and audit.

2.0 Scope

This guidance note applies to all Queensland coal mining operations under the *Coal Mining Safety and Health Act 1999* (Qld) (CMSHA), focusing on controlling and monitoring fumes generated from welding and other hot work activities, where these activities directly involve metals or metal alloys. Processes involving non-metal materials such as vulcanising rubber and plastics including polyethylene pipes are excluded from this guidance note.

For the purpose of this guidance note, hot work includes welding, brazing, soldering, heating, gouging, cutting and grinding, in which the surface temperature of the work, or a tool used for the work, is likely to exceed 150ºC.

This guidance note does not cover the other general safety hazards associated with welding and hot work, including:

- electrocution
- fire and explosion
- burns
- electric shock
- vision damage
- hearing loss including ototoxic effects
- exposure to intense ultraviolet (UV) radiation.

Guidance for managing these hazards should be sourced from other published documents and a list of additional resources including best practices for welding safety, has been provided in Appendix J.

General information only is provided about the toxic gases typically generated from arc welding processes, including ozone (O_3). At the time of writing this guidance note, there was limited data available from Queensland coal mines on

exposure monitoring results and control effectiveness for these gases. When more data becomes available, this guidance note will be updated.

3.0 Legislation

In addition to the objects of the CMSHA, section 6, this guidance note also aligns with Section 42 of CMSHA (State of Queensland 2024), which outlines the site senior executive's (SSE) obligations in relation to the safety and health of persons who may be affected by coal mining operations. These obligations include ensuring that risks are kept at an acceptable level and the development and implementation of a safety and health management system (SHMS). Airborne contaminants, such as welding fume, are a health hazard and exposure must be managed at an acceptable level.

Key obligations include:

- **Risk Control**: Ensuring risks from mining operations are at an acceptable level.
- **SHMS Implementation**: Developing and implementing a SHMS for all mine personnel, including contractors.
- **Training**: Ensuring CMWs receive training on hazards and risks related to their work.
- **Monitoring**: Providing regular monitoring and assessment of the working environment, procedures, equipment and installations.

Additional relevant sections of the Coal Mining Safety and Health Regulation 2017 (Qld) (CMSHR):

- **Section 46**: Employer must arrange for health assessment
    - An employer must ensure a health assessment is carried out for each person to be employed or employed, by the employer as a coal mine worker.
    - The health assessment must be carried out by, or under the supervision of the employer's appointed medical adviser.
- **Section 46A:** Content of health assessment
    - The health assessment must be carried out in accordance with the approved form.
- **Section 46B**: Other matters to be covered in health assessments
    - Health assessments may include matters not covered in the health assessment approved form if a risk assessment has been undertaken for a task and where the appointed medical adviser considers the person needs to be assessed to achieve an acceptable level of risk.
- **Section 49**: Monitoring for workers' exposure to hazards
    - The SHMS must provide for periodic monitoring of the level of risk from the hazard.
    - The SHMS must also provide for the notification of workers, and employers and the employer's appointed medical adviser of increased risk levels.
    - Ensuring periodic monitoring of workers' exposure to hazards.
- **Section 96 (a) and (c) (ii)**:

- A coal mine must have a standard operating procedure (SOP) to control access to, and work in, confined spaces at the mine.
- A SOP must exist that protects against hazards from cutting and welding, including in confined, wet, or dusty locations.

- **Section 102 (A) (c) and 149 (K):**
  - A surface mine and an underground mine must have a principal hazard management plan (PHMP) for airborne dust or any other airborne contaminants.
- **Section 263**:
  - Authorisation from the underground mine manager is required for hot work in underground mines, except in permanent underground workshops. The authorisation must be included in the mine record.

This guidance note is designed to assist coal mining operations to satisfy the requirements of CMSHA and CMSHR, in relation to managing welding fume.

4.0 Background

Welding is the process of joining two or more metal pieces together by applying intense heat, often with the use of consumable electrodes or filler materials. Welding fumes are generated due to the intense heat and primarily come from the base metals, electrode materials, and any surface contaminants or coatings on the metal being welded (Safework Australia 2024). These fumes consist of airborne particulates, ranging in size from 0.01 to 1 micrometre (µm), with inhalation being the primary exposure route (Australian Institute of Occupational Hygiene 2022). Fumes are also generated from other hot work processes associated with welding, including cutting and gouging.

In addition to particulates, gases such as oxides of nitrogen (NOx), carbon oxides, and ozone (O_3) are formed during arc welding.

4.1 Health effects from exposure

Inhalation of welding fume and gases is associated with a range of short and long-term health effects that can have serious health consequences (Figure 1). The most significant is the established link between welding fume exposure and lung cancer. Studies indicate that welders have a 30 to 40% higher risk of developing lung cancer compared to the general population and there is a probable association with kidney cancer (Figure 1) (Safe Work Australia n.d., Honaryar MK 2019). Other long-term health effects include occupational lung diseases such as occupational asthma, chronic obstructive pulmonary disease, emphysema, bronchitis and siderosis. Acute effects from exposure can also irritate the upper respiratory tract (nose and throat), and result in a flu like response known as metal fume fever and increase the risk of pneumonia.

Other health effects may occur as a result of exposure to specific compounds or metals within the fume. For example, exposures to welding fume containing nickel and chromium, can result in skin irritation or allergic contact dermatitis. Whilst exposure to welding fume containing lead or manganese may lead to neurological effects. Many metals commonly found within welding fume are known to be ototoxic, including lead and manganese. Ototoxic substances may damage the cochlea (inner ear) and/or the auditory nerve pathways to the brain, leading to hearing loss and

tinnitus. The combined effects of exposure, to welding fume and noise can have a synergistic effect, leading to higher levels of hearing impairment, than noise exposure alone. It is recommended that the noise exposure of workers (based on a daily shift) exposed to any ototoxic substances, be reduced below 80 dB(A) (8-hour equivalent) (Safework NSW 2024).

Chronic exposure to ozone may contribute to occupational asthma and may also exacerbate existing respiratory conditions including COPD. All open arc processes will generate ozone gas.

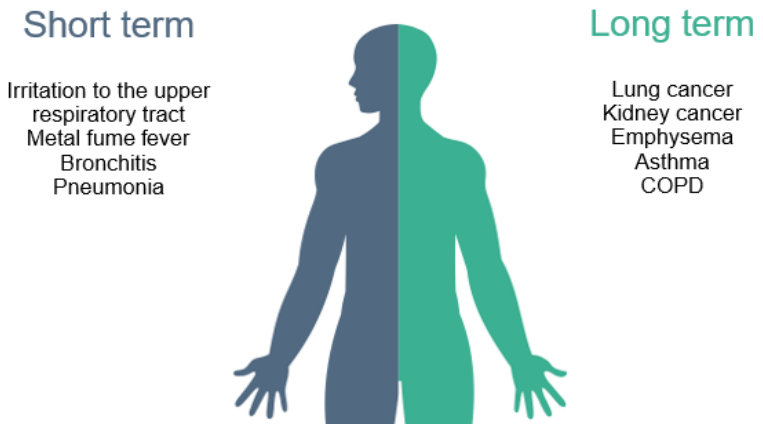


Figure 1: Health effects from exposure to welding fume

4.2 Generation of fume/gas from different types of welding and allied hot work processes

The fume cloud consists of particulate from the base metal, welding consumables and any residual surface coatings. The amount of fume produced, and the composition of the fume cloud will vary depending on the type of welding or hot work conducted, and the composition of consumables and the parent metal.

4.2.1 Welding fume emission

A detailed description of the welding fume composition that is generated by the common welding and hot work methods used in Queensland coal mines, has been provided in Appendix B. These welding methods include:

- Gas metal arc welding (GMAW)
- flux-cored arc welding (FCAW) (gas shielded and self-shielded)
- arc air gouging
- plasma cutting and gouging
- gas tungsten arc welding (GTAW)
- manual metal arc (MMAW) or stick
- thermal cutting and gouging
- brazing and soldering.

4.2.2 Fume emission rates for common welding/hot work methods

The relative fume emission rates for several of these common welding and hot work methods is summarised in Figure 2. This provides an indication of the different exposure potential, and the higher requirement for controls from different methods.

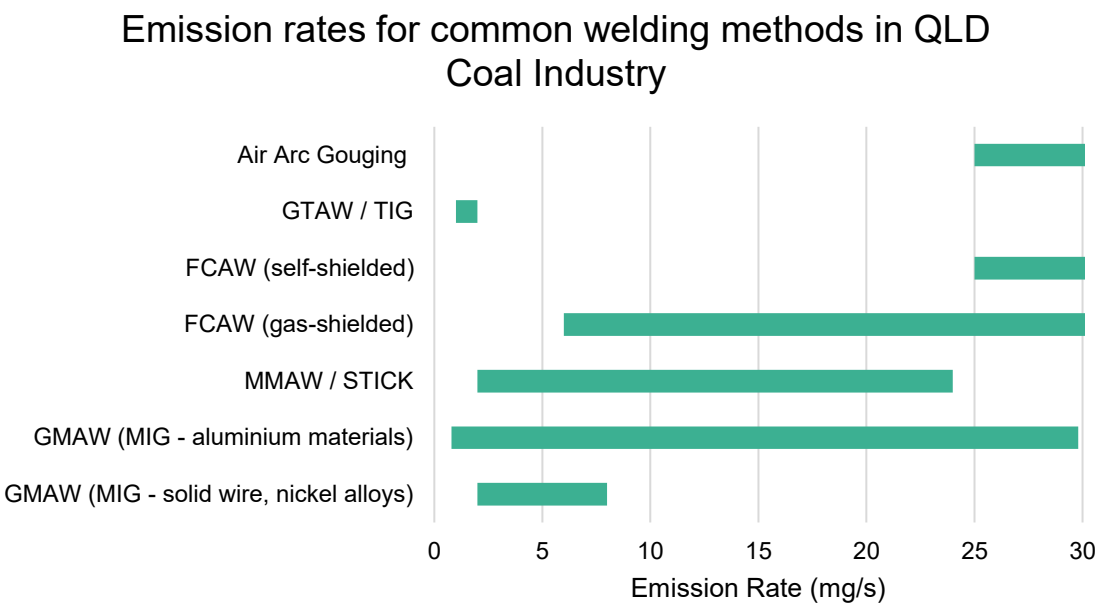


Figure 2: Types of hot work activity and fume emission rates, for common weld methods used in the Queensland coal industry. This figure was populated using emission rates from Welding Work: Technical Rules for Hazardous Substances (TRGS 528) (Committee on Hazardous Substances (AGS) 2021)

4.2.3 Types and composition of consumables

Welding consumables generally contain metals and elements that assist the welding process and protect the weld from the atmosphere. For example, some hard facing consumables often contain high levels of manganese. A list of various types of welding and their composition is included in Appendix D. Further composition detail for each product can be obtained from the consumable's safety data sheet.

4.2.4 Surface coatings

If a surface coating is present on the material being welded, the welding fume will contain oxides of the metallic coating. For example, a galvanised material will contain high levels of zinc and zinc oxides in the fume generated. In the case of a painted surface, the paint will decompose in the arc and a range of toxic polyaromatic hydrocarbons can be emitted within the welding fume. If the paint is an old lead-based paint, the fume will also contain lead.

In all cases, it is recommended for both personnel safety and weld quality reasons that all surface coatings and contaminants be removed prior to welding. There are various option for removing surface coatings including for example chemical stripping, or mechanical sanding and grinding. Consideration should be given to the secondary exposure risk when selecting a suitable option.

5.0 Exposure profile and welding methods used in Queensland coal mines

Most welding in the Queensland coal industry occurs in situ, typically during repairs and maintenance of heavy mobile plant and fixed structures at Coal Handling and Preparation Plants (CHPPs). Welding activities may also be conducted in workshops, confined spaces during plant shutdowns, or underground, though this is less common.

Most coal mines have designated welding bays either as standalone workshops or adjoined to or embedded within other larger workshops, typically used for mobile equipment maintenance. These vary significantly in terms of size, resources and set up. Some sites still have designated bucket shops and undertake significant amounts of intense welding and allied hot works processes in these facilities.

When designated welding areas are being designed consideration should be given to facilitating layout and installation of fixed and portable exhaust ventilation systems and, the separation of welding processes from other activities to minimise secondary exposure.

5.1 Boilermaker population and welding practices

A survey conducted by the coal inspectorate between 2022 and 2024 identified that the boilermaker workforce in Queensland's coal mines is approximately 56% full-time employees and 44% contractors (Figure 3). Welding practices differ between these groups, with contractors more likely to weld for longer hours each day, weld for a larger number of days each week and more likely use Metal Inert Gas (MIG) welding (Figure 6), which may result in differing exposure risk profiles.

The majority of welding tasks used on coal mining sites involve mild steel (Figure 4). Other specialty materials such as hard-facing, quench and tempered steels (e.g. bisalloy, weldox, hardox, roqlast or roqtuf), while often applied during the manufacturing of mining equipment, are used to a lesser extent on the mine site. Welding with stainless steel, aluminium and galvanised metal components is relatively uncommon.

FCAW and MIG are the most common welding methods employed, with GTAW (TIG) and stick welding used in more specific circumstances. (Figure 5).

A list of typical welding tasks performed at coal mine sites has been summarised in Appendix E.

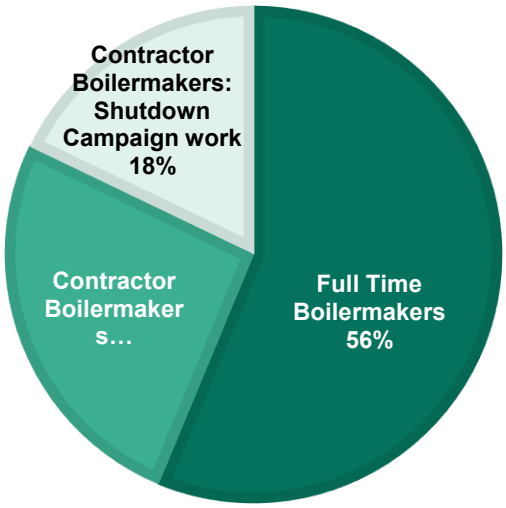


Figure 3: A breakdown of the welding population showing contractors and full time employees.

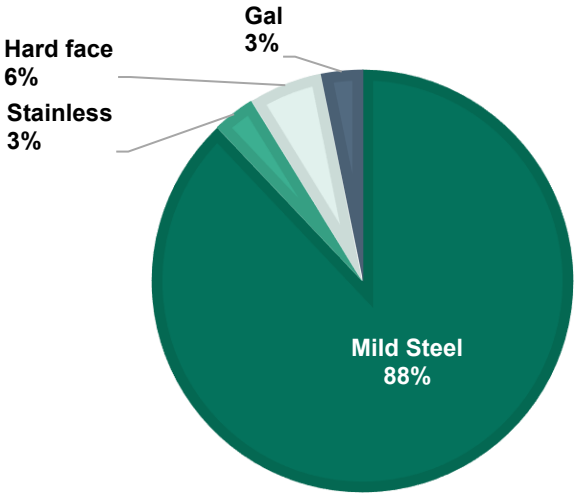


Figure 4: A breakdown of the welding materials used

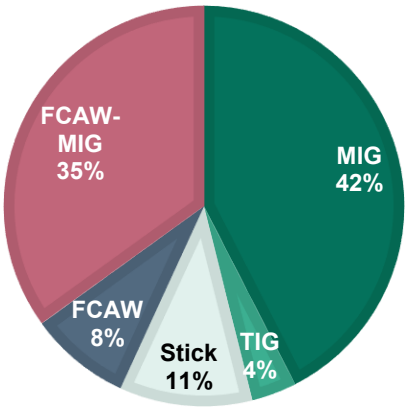


Figure 5: Common weld methods used in the Queensland coal mining industry**

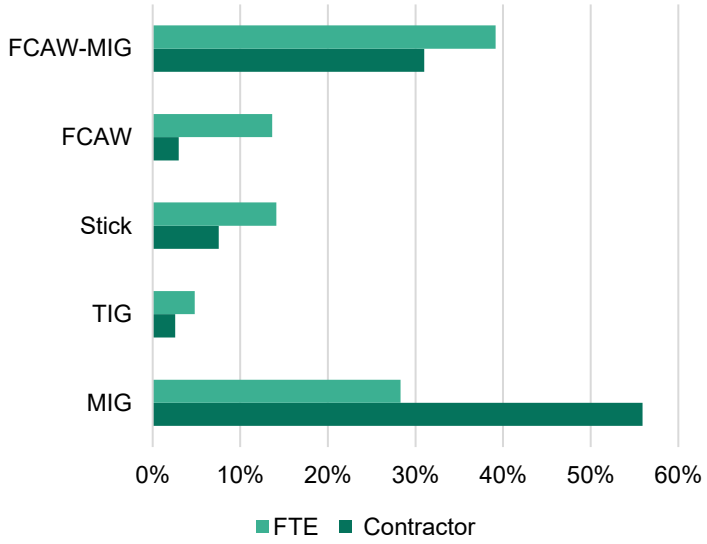


Figure 6: Welding methods used for Full Time Employees (FTE) vs Contractors **

**The figure presents data from a questionnaire where FCAW refers to self-shielded flux-cored arc welding, and FCAW-MIG refers to gas-shielded flux-cored arc welding. MIG indicates the use of solid wire. Caution should be applied with interpreting this data, as there may have been some confusion regarding these terms due to wording used in the questionnaire.

5.2 Welding fume exposure

Personal exposure to welding fume is routinely monitored at Queensland coal mines. These samples are collected using inhalable size selective devices positioned in the boilermaker's breathing zone, although outside of any respiratory protection equipment (RPE), and therefore represent a measure of unprotected exposure. The rates of monitoring have increased significantly over the last 5 to 10 years as awareness of the health risk has been

communicated. Samples collected for total welding fume are typically analysed for individual metal components and compared against the relevant Occupational Exposure Limit (OEL) (Safework Australia 2024). A reduction in the OEL for total welding fume from 5 mg/m3 to 1 mg/m3 was introduced by Safe Work Australia (SWA) and came into effect in early 2024.

Currently there is a combination of full shift and task based (episode) monitoring being conducted at coal mines. Full shift personal exposure monitoring conducted on coal mines over the last decade indicates that the current OEL for total welding fume of 1 mg/m^3 has been consistently exceeded. Currently the CMSHA only prescribes a limit for total welding fume in schedule 6: General body concentrations for atmospheric contaminants of 5 mg/m3 This only applies to the underground coal mine environment. There is no limit prescribed for surface coal mines or the surfaces parts of underground coal mines. In these instances, where there is no prescribed limit, the limit published by SWA (Safework Australia 2024) has generally been applied as the default.

Each year Resources Safety Health Queensland (RSHQ) requests coal mines provide records of all personal monitoring for total welding fume, including the individual metal components. RSHQ now holds a dataset comprising of approximately 2,400 valid personal welding fume exposure records. This includes full shift (70% of total) and task based (30% of total) monitoring results with records dating back to 2009.

Figure 7 shows the average total welding fume results from personal exposure samples collected for the boilermaker Similar Exposure Group (SEG) at various surface coal mines over the last five years. Each colour represents a different mine. The average is calculated across a set of generally 6 to 10 samples for each mine.

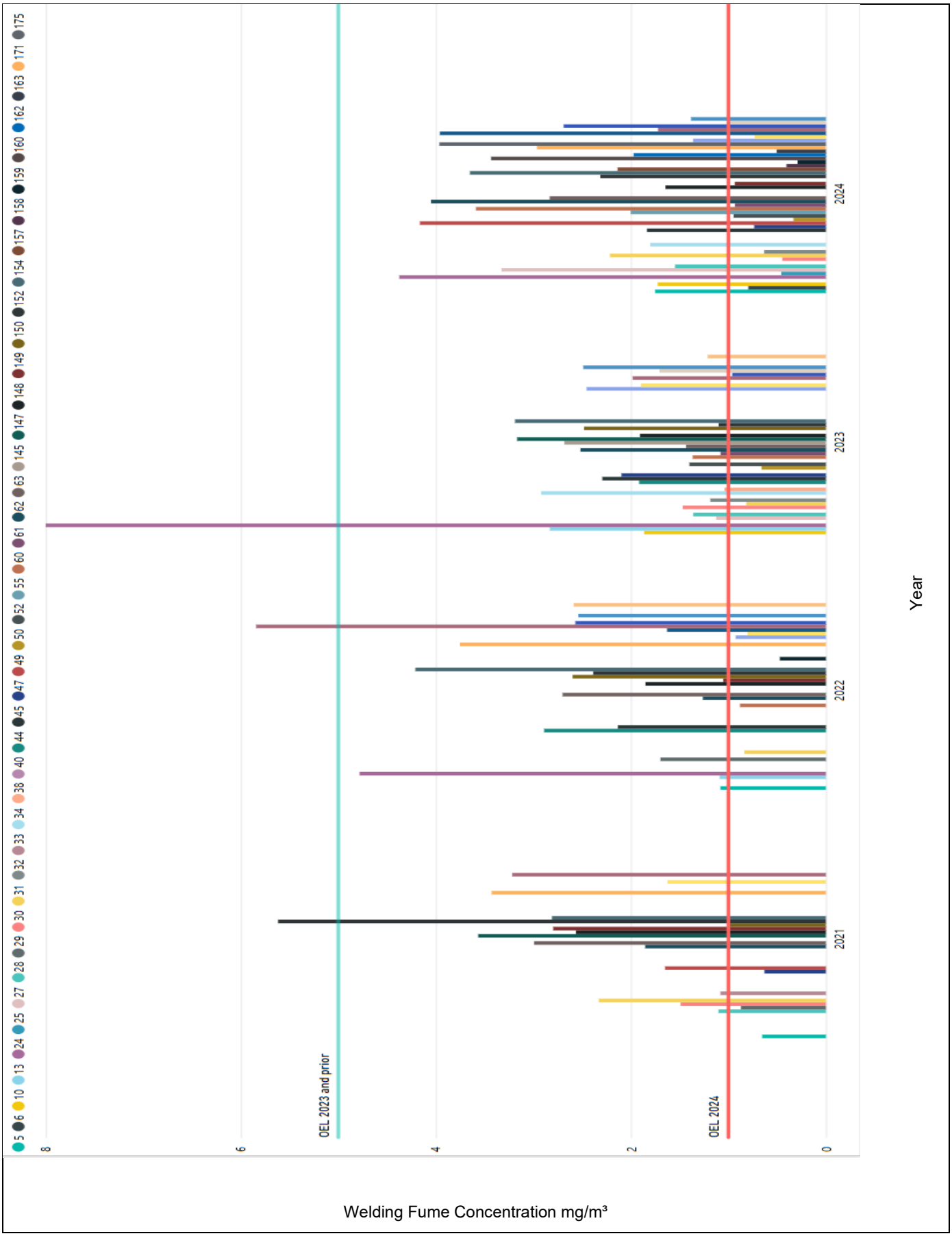


Figure 7: Total welding fume average exposures for boilermakers (includes boilermaker on the surface at both underground sites and surface sites, full shift monitoring only). Each colour represents a different mine site.

In 2024, 286 samples were reported to RSHQ, from 50 sites (80% of all Queensland coal mines). The data indicates workers are continuing to be exposed to high levels of welding fume and self-reporting that their RPE is not being worn.

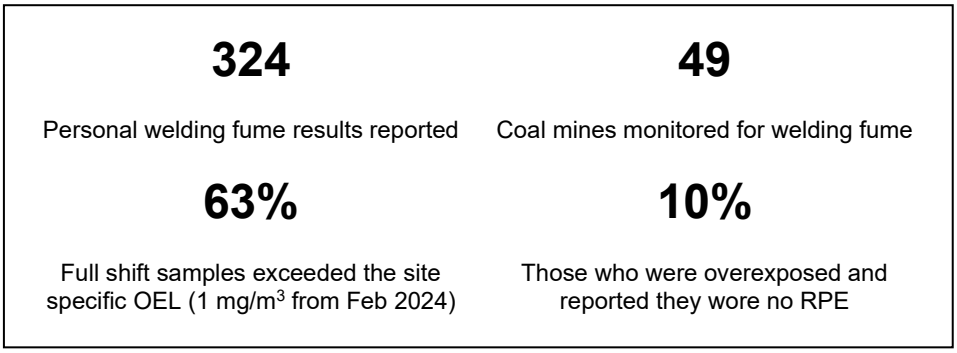


Figure 8: Number of samples, percentage of exceedances and workers exposed to welding fume.

6.0 Risk management approach for welding fume

6.1 Risk management process

The potential health effects from welding fume exposure and the level of exposure from the methods typically applied within Queensland coal mines, has been well established in other sections of this guidance note. The primary purpose of conducting a risk assessment within the context of this guidance note, is to identify and rank the potentially exposed workers and SEGs within the mine and determine the level of monitoring and controls to be applied at the mine.

The primary consequence of interest from welding fume exposure in this context are the chronic health effects, including lung disease and in particular lung cancer. The nature of these chronic health effects is such that the risk of harm is related to cumulative exposure over many years and long latency periods, and the symptoms and diagnosis of disease may be 10 to 30 years after exposure. A different approach is required when assessing and ranking these risks, as compared with the methods applied to assess acute safety risks, where the impacts occur at the time of the incident or shortly after. The International Council on Mining and Metals (ICMM) have proposed a separate Health Risk Assessment (HRA) approach, that includes a number of options to rank risk according to exposure level and this approach has been adopted for reference in this guidance note.

Risk management is a systematic process to eliminate or minimise the potential for harm to people. Regarding welding fume, this process includes the following steps:

- **Identify** which workers are at risk of exposure to welding fume, including boilermakers and those working in the immediate area.
- **Assess the risk** using exposure estimates to understand the nature of harm that could be caused, and determine what sources and processes are causing that risk.

- **Control risks** if it is not reasonably practicable to eliminate risk, implement the most effective control measures that are reasonably practicable in accordance with the hierarchy of controls.
- **Review and monitor** control measures to ensure they are working as planned.

Figure 9 outlines the risk management process steps applied within this guideline.

Risk assessment must involve appropriate consultation between relevant parties including an occupational hygienist and a cross section of workers, including boilermakers and others potentially affected by welding fume hazards.

Each mine must identify and assess exposure to welding fume as part of the mine operations HRA. As with all other identified hazards, the procedures documented within the mine's SHMS (e.g. PHMPs, Workplace Risk Assessment and Controls (WRACs), SOPs need to incorporate the outcomes identified in the HRA. The HRA should be reviewed and updated within 5-yearly intervals. Or earlier if there is a significant change to welding practices, exposure, and/or there is a change in the OELs.

A summary of the risk management process is outlined in the flow chart below (Figure 9).

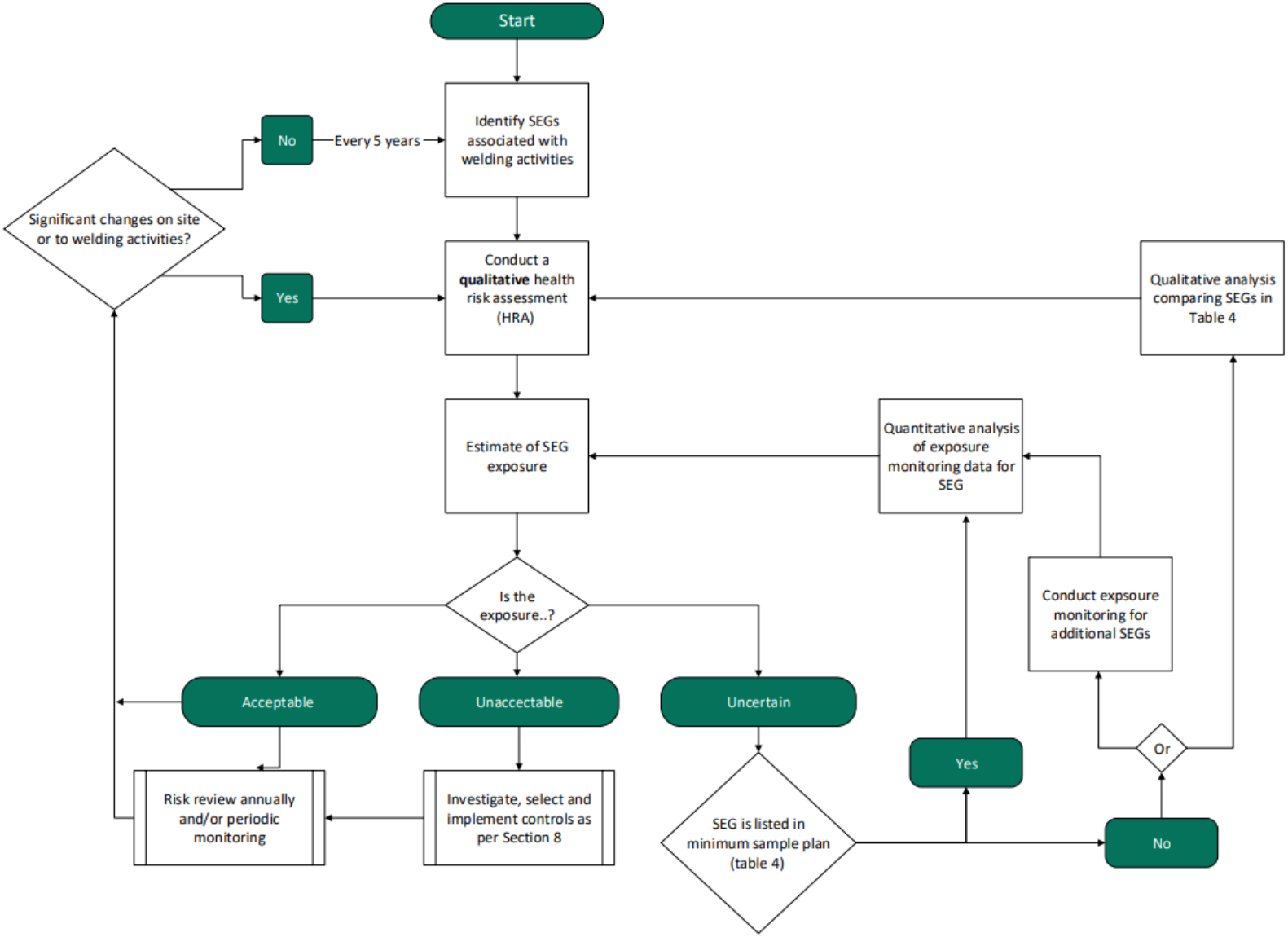


Figure 9: Risk Management Process Steps

6.1.1 Identifying the hazard

The purpose of hazard identification within the context of this guidance is to identify and rank different exposure groups and activities involved with applying hot work methods. The first step in the risk management process is to identify all hazards associated with the welding process. Consideration should be given to identifying the following at the mine site:

- All SEGs engaged in welding tasks and other hot works (RSHQ n.d.).
- All exposed workers within SEGs who may be exposed to welding fume (including boilermakers, firewatchers, trades assistants and others within the immediate working area).
- Welding methods and hot works being conducted.
- Materials and consumables being used.
- Work location and environment (indoors, outdoors, workshops).
- Frequency (i.e. days per week) and duration (hours per day) these hot work methods are applied by each group.

A list of common welding tasks conducted on Queensland coal mine sites is provided in Appendix E.

The hazard identification process should involve the following steps:

- Desktop review of previous HRAs, previous monitoring surveys and exceedance investigations, and safety data sheets.
- Walk through survey - this may include a questionnaire to record specific weld methods, consumable types and controls used by different groups. This should include discussion with boilermakers and supervisors.
- Ranking hazards - this may involve ordering the hot work/welding methods used by each group according to fume emission rates and potential to cause harm. The fume emission rates for different weld methods is summarised in Figure 2. An example of a hazard rating criteria is shown below (Table 1 (ICMM 2016)).The hazard rating for most welding activities where exposures to welding fume are estimated to exceed the OEL and likely cause adverse health effects, will be medium or high, depending on duration/frequency of exposure.
- Recording what welding fume controls are in place and what level in the hierarchy of controls (Figure 10). This should include whether the welding fume controls are being used as intended, if they are being inspected and maintained, and is there a system for monitoring control status and integrity.

Table 1: Example criteria to rate hazards adapted from ICMM Good practice guidance on occupational health risk assessment (ICMM 2016)

Hazard Rating	Definition
1 – Very Low (Minor health effects)	Exposure is unlikely to lead to harm.
2 – Low (Reversible health effects)	Non-life-threatening reversible health effects e.g.: irritation to upper respiratory tract, metal fume fever, bronchitis or pneumonia.
3 – Medium (Adverse health effects)	Adverse health effects that are permanent but do not significantly affect quality of life or longevity. Health effect that may be mildly limiting or disabling and therefore could lead to a change of occupation and lifestyle.
4 – High (Significant and severe health effects)	Adverse health effects that are generally permanent and could lead to a significant reduction in quality of life and/or longevity. Continued exposure is generally likely to lead to permanent physical or mental disability or a long-term limiting illness e.g. kidney or lung cancer.

While the focus of this guidance note relates to the exposure to welding fume, a risk management process should also be applied to the other general safety hazards associated with welding and hot work activities. Guidance for managing these hazards should be sourced from other published documents and a list of references including best practices for welding safety have been provided Appendix J.

6.1.2 Assessing exposure to welding fume

After completing the hazard identification phase, qualitative and quantitative risk assessment methods should be applied to estimate the incidence and severity of health effects, as a result of workplace exposures. This is then used to develop and prioritise controls and is typically done without considering controls in the first instance. Qualitative and quantitative risk assessment of welding fume should involve consultation with a certified occupational hygienist (COH).

The health effects of interest from welding fume exposure are chronic lung diseases and in particular, lung cancer. The probability should estimate the likelihood of exposure above levels known to cause health effects.

Qualitative risk assessment

Qualitative risk assessment is conducted as a first step to estimate whether exposure is uncertain and further monitoring is required to understand the risk, or if exposure is unacceptable and additional controls are required. This should then be used to inform the broadbrush risk assessment and to identify significant health risks that require more comprehensive measurement and control. This assessment assists in determining:

- the extent of the risk (degree of exposure and potential health effects)
- the effectiveness of current control measures, considering the hierarchy of controls (refer section 8)
- actions to be taken for additional controls and monitoring.

Quantitative risk assessment

Once the qualitative risk assessment has been completed and monitoring conducted to determine the levels of exposures within priority SEGs quantitative risk assessment should be used on an ongoing basis to determine the levels of risk, effectiveness of controls and priorities for further control investigation. An example of quantitative exposure rating criteria is shown below in Table 2.

Semi-quantitative risk assessment should be completed for any SEG groups that are not included in the list of priority SEGs selected for minimum exposure monitoring requirements (Table 4). This requires professional judgement to estimate the relative exposure level for each SEG that has not been monitored, by comparing the frequency and duration of hot work activities, and weld methods with those SEGs that have been monitored. (EN 689 2019)

Table 2: Quantitative Exposure Rating System

Rating	Exposure Band	Definition
Low	Less than 50% of OEL	Average exposure estimate for a SEG is less than 50% of the OEL. Exposures are at or well controlled to below the OEL, there are less likely to be exceedances of the OEL and this level of exposure is likely to cause little or no adverse health effect.
Medium	Between 50% and 100% of OEL	Average exposure estimate for a SEG exceeds 50% of the OEL but is less than 100% of the OEL. Exposures are at or controlled up to the OEL, there is a potential for exceedances of the OEL and this may cause an adverse health effect in some workers, including vulnerable individuals.
High	Greater than OEL	Average exposure estimate for a SEG exceeds 100% of the OEL. Exposures are above and/or not controlled to below the OEL and are likely to cause adverse health effects in the majority of workers exposed, either in the short or long term.

Risk acceptability

If the average exposure for a SEG is above the OEL, then the risk is deemed to be unacceptable, and further controls should be implemented. In addition, when considering risk acceptability, controls should be implemented to ensure the exposure risk is controlled to as low as reasonably achievable. This includes the application of higher order controls before considering any protection provided by RPE.

When assessing risk, the statutory exposure limits prescribed for airborne contaminants within the CMSHR legislation, shall be applied in the first instance. In the absence of statutory limits being prescribed for specific airborne contaminants, the OEL values published by Safework Australia should be applied.

6.1.3 Controlling the risk

Controls implemented to manage welding fume exposure should follow the hierarchy of controls. This should consider eliminating or minimising the level of fume by focusing on the process first, followed by ventilation and

finally RPE. In most cases a combination of controls will be required in order to achieve an acceptable level of risk.

A detailed list of practical welding fume controls that follow the control hierarchy and which are available for use in the coal mining industry, have been outlined in Section 0 and are summarised in Appendix E.

6.1.4 Review and verification of control effectiveness

To ensure welding fume exposure remains effectively controlled, the risk management process should incorporate a documented process to (i) validate the effectiveness of new controls introduced; (ii) periodically review the effectiveness of existing welding fume controls; and (iii) include a process for routine in-shift validation of existing control performance.

(i) The effectiveness of new controls can be validated by taking measurements before and after new fume controls are introduced at the mine, either by using gravimetric or real time sampling methods. Alternatively, the results of validation measurements reported from other sites, or published by supplier/manufacturer groups or by research studies, for the same control type, may be used for reference.

(ii) Periodically reviewing the effectiveness of existing welding fume controls should include an annual review of the exposure monitoring results from the mine, maintenance inspection reports completed on fume control equipment, and any incident reports in relation weld fume control devices.

(iii) Routine in-shift validation of fume control performance should include in-shift inspection and monitoring the operating condition of the device and comparing this with design parameters. For example, this may include direct measurement or checking display gauge readings for pressure, flow rate, or air velocity.

(iv) Manufacturers of fume controls should facilitate in-shift performance validation, by installing monitoring and display systems onto the equipment, or by providing measurement tools, to confirm design parameters are being maintained during equipment operation. For example a flow or pressure gauge mounted onto a fixed exhaust ventilation system to indicate the system is functioning as intended.

6.1.5 Audit and review

A review of the risk assessment should be conducted when controls fail to reduce exposure to an acceptable level, before any workplace or process changes, when new hazards are identified, when consultation with CMWs or other stakeholders highlight the need for review, or at the request of a Site Safety and Health Representative.

An audit and review of procedures documented within the mine's SHMS, in relation to the monitoring and control of welding fume, should be conducted periodically to ensure ongoing effectiveness and continuous improvement, and exposure to welding fume are being managed to an acceptable level of risk.

6.1.6 Change management

The change management process should be triggered when there is a proposed change to the type of hot work and welding methods, consumable types and/or materials being welded, at the mine. The change management

process should identify the potential impacts on welding fume exposure and the potential introduction of new hazards and risks.

Before new control measures are applied to manage exposure to weld fume, a change management process should be applied, to ensure the introduction of any new risks are identified and managed effectively. For example, a poorly controlled fume ventilation system may disturb the gas shield around the molten metal pool, resulting in reduced weld integrity.

7.0 Personal monitoring for welding fume exposure

The monitoring of welding fume exposure should be generally aligned to the concepts and requirements for monitoring respirable dust, as specified in Recognised Standard 14: Monitoring respirable dust in coal mines (RS14).

Where there are unique requirements for monitoring welding fume these are outlined in the guidance below.

Where welding fume has been identified in the mine's PHMP for airborne contaminants, this should include requirements for a welding fume personal exposure monitoring program.

7.1 SEGs

The SEGs relevant to welding activities at each mine must be established and used to develop the mine's welding fume monitoring program.

The list of SEGs applied to the monitoring and reporting of sampling results for respirable coal dust in accordance with RS14, do not adequately define the smaller exposure groups of CMWs performing welding activities.

Therefore, a separate list of SEGs that perform common welding activities and which contribute to the majority of welding fume exposure, within coal mining have been included in Table 3. For the purpose of reporting welding fume exposure monitoring data to RSHQ, all data should be assigned to the SEGs listed in Table 3. These are unique to welding activities and there is currently no requirement to monitor these SEGs for other airborne contaminants such as respirable dust and respirable crystalline silica.

The separate SEGs that perform welding activities have unique exposures not experienced by other workers and were identified from a review of industry data and surveys conducted in the development of this guidance note. It was also identified that some work groups with elevated exposure to welding fume were not being monitored. For example, historical monitoring for welding fume was typically focused on boilermakers performing intermittent breakdown repairs from the mobile equipment workshop, whereas more frequent and longer duration welding exposures were being experienced by groups of CMWs doing refurbishment and shutdown work on heavy mobile equipment (HME), and these were not being monitored.

Notwithstanding the requirements to use SEGs listed in Table 3 for reporting purposes, mines should establish and use their own SEGs based on observations and the exposure profile of their specific work groups, and cover the activities listed below:

- Plant maintenance;
    - heavy mobile equipment maintenance
    - infrastructure maintenance (e.g. buildings, conveyors, run-of-mine (ROM) bins)
    - coal processing plant maintenance.
- Maintenance locations;
    - workshop based
    - field based.
- Maintenance type;
    - breakdown
    - routine minor plant shutdown
    - major plant shutdown/overhaul
    - refurbishment projects (e.g. buckets, truck trays).

Table 3: SEGs applicable to welding

SEG	CODE	SUB GROUP	PRIMARY TASK DESCRIPTIONS
Underground mine site			
Boilermaker Underground	**QCU030**		CMWs conducting welding/hot work tasks underground.
Boilermaker Surface Workshop	**QCU031**		CMWs conducting welding/hot work tasks at the surface workshop of an underground mine. Includes repair/fabrication to mobile plant and ancillary equipment. Welding for intermittent periods during the shift.
Boilermaker Surface Infrastructure	**QCU032**		CMWs conducting welding/hot work tasks in situ, on infrastructure and fixed plant, on the surface of an underground mine. Includes repair/fabrication of fixed plant (e.g. ROM, conveyors, transfer towers) and involving both breakdown and planned maintenance.
Surface mine site			
Boilermaker HME Workshop	**QCS034**	Planned Maintenance	CMWs conducting welding/hot work tasks in and/or around the workshop. Includes repair/fabrication of mobile plant where this work is planned and scheduled in advance and typically involves prolonged welding over significant portion of the shift.
		Unplanned/Breakdown	CMWs conducting welding/hot work tasks in and/or around the workshop. Includes repair/fabrication of mobile plant where this work is ad hoc and unplanned or in response to plant breakdown and typically involves intermittent welding for short periods of the shift.
Boilermaker Field Maintenance	**QCS035**	Planned Shutdown	CMWs conducting welding/hot work tasks in the field and away from the workshop. Includes repair/fabrication of mobile and/or fixed plant in situ and where the work is part of a planned shutdown. Typically involves prolonged welding over a significant part of the shift.
		Unplanned/Breakdown	CMWs conducting welding/hot work tasks in the field and away from the workshop. Includes repair/fabrication of mobile and/or fixed plant in situ and where the work is in ad hoc and unplanned or in response to plant breakdown. Typically involves intermittent welding for short periods of the shift.
Boilermaker HME Refurbishment Maintenance	**QCS036**	Planned Project works	CMWs conducting welding/hot work tasks in and/or around the workshop, or in the field, where this involves major overhaul/refurbishment of large mobile plant components (e.g. buckets, trays, dragline tub). Includes repair/fabrication being continuously applied to the same plant component for the full shift duration and repeated over multiple shifts.
Coal handling preparation plant and ROM			
Boilermaker CHPP Fixed Plant Field Maintenance	**QCP009**	Planned Work	CMWs conducting welding/hot work tasks in the field and away from the workshop. Includes repair/fabrication conducted in situ and where this is part of planned campaign maintenance work or shutdown activities directed at fixed plant components (e.g. structural repairs, handrails, gridmesh, chutes). Typically involves welding over a significant part of

SEG	CODE	SUB GROUP	PRIMARY TASK DESCRIPTIONS
			the shift and repeated over multiple shifts. Excludes unplanned breakdown work.
		Unplanned/Breakdown	CMWs conducting welding/hot work tasks in the field and away from the workshop. Includes repair/fabrication conducted in situ and where this work is ad hoc and unplanned or in response to plant breakdown. Typically involves intermittent welding for short periods of the shift. Excludes planned maintenance work.
Boilermaker CHPP Workshop	**QCP010**	Planned Maintenance	CMWs conducting welding/hot works tasks in and/or around the CHPP workshop.
		Unplanned/Breakdown	Includes repair/fabrication of plant where this work is planned and scheduled in advance and typically involves prolonged welding over significant portion of the shift.

7.2 Minimum sampling requirements

The review of industry data and surveys conducted in the development of this guidance note, has confirmed that weld fume exposure in Queensland coal mines is characterised by a relatively consistent and limited range of weld methods, materials, and consumables. The primary difference in exposure risk between groups of CMWs, is the frequency and duration of fume generating activities. This can be broadly separated into two exposure categories:

- Prolonged welding periods (e.g. > 6 hours), typically scheduled in advance (e.g. HME plant maintenance shutdowns).
- Intermittent welding periods (e.g. ≤1–2 hours), typically unscheduled (e.g. HME plant breakdowns).

7.2.1 Minimum samples and frequency of monitoring

An exposure monitoring plan has been developed to characterise welding fume exposures to workers within the Queensland coal mining industry (Table 4).

This minimum sampling plan has been developed to effectively sample and characterise the exposure profiles of a limited number of selected SEGs. These SEGs represent the range of potential welding fume exposures from both prolonged and intermittent welding categories and most common welding tasks applied within Queensland coal mines.

The number of samples specified for each SEG is the minimum number required to support simple statistical analysis to estimate the exposure for that group (S.M and Selvin 1987).

A COH or equivalent should develop and endorse the monitoring program and conduct annual reviews of the program.

Table 4: Minimum sampling plan for Welding SEGs

Boilermaker SEGs	Boilermaker HME Workshop QCS034	Boilermaker HME Refurbishment Maintenance QCS036	Boilermaker CHPP Fixed Plant Field Maintenance QCP009	Boilermaker Surface Workshop QCU031
Sub Group	Planned Maintenance	Planned Project works	Planned works	
Recommended Sample Type	Task Based+	Full Shift	Full Shift	Task Based+
Min Sample numbers	3^	6*	6*	3^
Sample period	Annual	Annual	Annual	Annual

+	Refer section 7.3.2 for additional information.
*	Rappaport, Selvin, Roach 1987 - 6 samples based on >3.5 Geometric Standard Deviation and F (mean to OEL)=3.0
^	Minimum of three (3) samples to be collected annually to enable analysis of 6 samples in total over two years.

7.2.2 Limitations with minimum sampling plan

This minimum sampling approach has been chosen solely for estimating welding fume exposure for the purpose of this guidance note. The minimum sampling numbers have been applied so as not to compromise the sampling capacity already committed to the previously established monitoring programs for other airborne contaminants, including but not limited to respirable dust and crystalline silica.

As a consequence of this approach, there are a number of limitations inherent within the minimum sampling plan and these should be considered when interpreting the results of monitoring over time.

Limitations with minimum sampling plan include:

- Exposure monitoring for welding fume is inherently biased in that sampling is typically only conducted when welding activities are being performed and do not include periods or shifts when there is no exposure. Consequently, the result of this sampling only provides an estimate of exposure intensity during welding activities and this may overestimate the actual long-term exposure.
- Full-shift sampling is used for groups engaged in prolonged fume-generating activities. These results can be compared directly with OELs. Cross-contamination can occur from airborne contaminants generated by non-welding tasks that occur during the sampling period.
- Task-based sampling is used where exposures are intermittent and short duration. While it reduces cross-contamination uncertainty, the sample period may not be representative of exposure over the whole shift and therefore introduces worst-case sample bias. These results cannot be directly compared with OELs without applying professional judgement and having comprehensive knowledge about what exposures occurred during portions of the shift that were not monitored.
- Focusing sampling on a limited number of welding SEGs improves the reliability of estimating exposure for the groups being monitored. This approach does not sample all welding SEGs and estimating the

exposure for these groups will require qualitative analysis using extrapolation. This will result in limited data to assess the effectiveness of controls being applied by these groups.

- The limited numbers of samples for each welding SEG collected each year, may result in variable results, and this should be considered when interpreting results and estimating welding SEG exposure. The review of exposure data over a greater period than one year may be required where fewer than six samples have been collected.
- While priority should be given to monitoring the welding SEG specified in Table 4, adjustments may need to be made for some mines where the population of boilermakers and/or allocation of welding work, does not align with these groups. For example:
    - For small mines where welding tasks may be infrequent and insufficient to be allocated into the specified welding SEG, monitoring for welding fume should be conducted on welding groups applicable to the mine.
    - For large mines where the welding SEG listed in Table 4 do not adequately cover the range of welding tasks and potential exposures at the mine, additional sampling should be conducted to characterise the exposure profile of these other groups and/or larger populations of exposed CMWs.
    - Additional welding SEG may be monitored at the discretion of the mine site, and this could be a requirement resulting from risk assessments, investigations or concerns raised by CMWs.

7.3 Personal exposure sampling

Sampling should be scheduled with maintenance planners and coordinators to ensure plant shutdowns and other project work is included in the monitoring plan. Sampling should primarily focus on welding tasks where practical, although other hot work tasks directly associated with welding may be included. Workers should be randomly selected for each sampling period to minimise sampling bias and samples should be representative of exposure to welding fume.

7.3.1 Competency requirements

The competencies required to develop exposure sampling plans and conduct personal exposure sampling should align with the requirements applied for respirable dust in section 4.3 of RS14. In addition, a person who is developing sampling plans and conducting personal exposure sampling for welding fume, should have familiarity with this guidance note, and the following:

- welding methods
- potential sources of sample contamination/damage/invalid samples
- welding fume controls
- safety hazards associated with welding and hot works.

7.3.2 Sampling methods

Personal exposure sampling

Samples must be collected from the breathing zone in accordance with the standard sampling method (ISO

10882-1:2024). For the purpose of assessing exposure within the context of this guidance note, sampling heads should be positioned in ambient air behind face shields but outside any respiratory protection. For example, given that Powered Air-Purifying Respirator (PAPR) provides both respiratory protection and incorporates the face shield, the sampling head must be placed in the breathing zone outside the PAPR helmet.

Where the boilermaker is wearing a face shield (that does not incorporate a respirator) the sampler must be positioned in the breathing zone behind the face shield. The sampler may be positioned and attached to the inside of the face shield. Alternatively, the sampler may be positioned outside and behind the face shield, by attaching the sampler high up onto the wearer's shirt collar.

In both cases the sampler must be orientated in accordance with manufacturer instructions, to ensure the size selective sampling of airborne exposure concentrations is conducted effectively.

The position of the sampler must be recorded on the field sheet and accompany the reporting of results.

The sampler may be positioned in other locations within the breathing zone for the purpose of verifying control effectiveness and not assessing exposure.

The total duration of all welding tasks and the type of welding method, across the full shift period, must be recorded on the field sheet (Appendix C). (EN 689 2019)

Task-based sampling

Samples should be collected only during welding fume generation activities. Multiple periods may be combined into a composite sample. All welding tasks conducted during the shift should be sampled. The sampling of non-welding activities should be minimised to avoid contamination and these recorded where they occur. A minimum runtime of 50 minutes per sample[1] is recommended unless shorter durations are justified by high-exposure tasks.

Full-shift sampling

Samples should cover at least 80% of the shift. This may include all types of hot work activities conducted during the shift.

7.3.3 Sampling equipment

Size selective sampling devices should be used to sample personal exposure to welding fume for comparison with the OEL. These devices must be certified by an accredited third party to conform with the standards for the relevant size fraction sampling convention (EN 481-1993) (BS EN 481 1993) (BS EN 13205-1,2:2014) (ISO 7709-1995 (ISO 2023)).

Equipment used to sample welding fume must be fit for purpose and maintained to achieve the objectives stated in the relevant sampling standards AS3640/AS2985 and this guidance note. Sampling equipment must be subjected to periodic calibration and performance checking, at least annually.

[1] This runtime has been established based on the calculation in clause 9.1.3.1 of ISO 10882-1 (ISO 10882-1:2024 2024), using a limit of reporting of 0.01 mg/m^3 for undifferentiated welding fume in the inhalable fraction.

Table 5: Sampling Standards and Equipment Requirements.

Equipment	Sampling standards/requirements	Examples of sampling equipment
Sampling pumps	ISO13137 Calibration to ISO 13137 – annex C - C.1 to C.4 (user tests for pumps), clause 7.10 and clause 7.11 (timer accuracy). Pumps must be tested to the specific application and with flow rates commonly operated.	
Flow meters	AS3640/AS2985 - Appendix B	
Sample heads – inhalable	AS3640 - Clause 6.2 (Standards Australia n.d.)	
Sample heads – respirable	AS2985 (Standards Australia 2009) Respirable dust size selective inlet to ISO 7708.	
Sample heads – hexavalent chromium	OSHA ID-215(V2) (Occupational Safety and Health Administration n.d.) 37mm closed face cassette in universal cassette holder.	

Additional requirements regarding calibration, performance checking, maintenance, and use of electrical equipment should align with section 5.1 of RS14.

7.3.4 Standard for collecting and analysing welding fume samples

All samples collected for the purpose of personal exposure assessment should be collected and analysed in accordance with ISO 10882-1:2024 (ISO 10882-1:2024 2024). Laboratories performing analysis of welding fume samples must be accredited by the National Association of Testing Authorities (NATA) to Australian Standards (AS) ISO/IEC17025:2018 general requirements for the competence of testing and calibration laboratories for the gravimetric determination of inhalable dust and respirable dust (AS ISO/IEC 17025:2018 International Organization for Standardization 2018). Analysis for welding fume metals must be in accordance with a standard method. All analytical results reported by the laboratory for personal exposure sampling, should include the uncertainty in the reported results.

As a minimum, the following sampling and analysis should be conducted:

- total welding fume (inhalable fraction)
- metal suite analysis (inhalable fraction).To include the following recommended minimum metals:

- Iron Oxide Fume (Fe), Manganese Fume (Mn), Nickel Metal (Ni), Total Chromium (Cr), Copper Fume (Cu), Zinc Oxide Fume (Zn).

Additional sampling and analysis should be conducted where other contaminants are identified and likely to contribute to an exposure risk, including but not limited to:

- Manganese (respirable fraction) – if the manganese (inhalable fraction) exceeds 50% of the shift adjusted OEL, consider additional sampling the respirable fraction.
- Hexavalent Chromium (inhalable fraction) - if the total chromium (inhalable fraction) exceeds 50% of the shift adjusted OEL, consider additional sampling for Hexavalent Chromium.
- Other contaminants, where these are considered likely to contribute to exposure based on the consumable and welding technique (refer Appendix D).

Analysis for total welding fume and metals should be in accordance with a recognised method. Examples include but are not limited to the following:

- Total welding fume - Gravimetric analysis to determine the total inhalable fraction mass of welding fume according to AS 3640.
- Metal suite analysis - Inductively coupled plasma optical emission spectroscopy (ICP-OES) or mass spectrometry (ICP-MS) is used after acid digestion of the filter media according to NIOSH 7300, 7301, 7303, or ISO 15202. X-ray fluorescence (XRF) may also be used.
- Respirable manganese - Gravimetric analysis to determine the respirable fraction mass of welding fume according to AS 2985, with manganese analysis included in the metal suite analysis.
- Hexavalent Chromium – Analysis of fume collected onto filter media using ion chromatography and UV detection according to OSHA ID-215 (version 2) (Occupational Safety and Health Administration n.d.).

Invalidating samples

Personal exposure samples that do not meet relevant method for the sampling and analysis of welding fume and metals are invalid and must not be used for estimating exposure. The criteria for which samples are classified as invalid should be in accordance with section 7.9 of RS14.

Limit of reporting and censored data

Personal exposure sample results that are reported as below the laboratory limit of reporting are referred to as censored data or non-detects and treatment of these should follow the requirements outlined in Section 7.4 of RS14.

Tampering with samples or results

A person must not tamper, or allow another person to tamper, with a sample or the results of a sample taken to monitor a worker's exposure to welding fume at a coal mine. This is further outlined in section 10 of RS14.

Sampling records

Records of personal exposure sampling performed for welding fume must be kept for 30 years at the coal mine and follow the requirements outlined in section 7.8.3 of RS14. Observations of the welding task(s) performed

and controls used during the monitoring period(s) should be recorded with sample records. The minimum information to be recorded for each sample is outlined in Appendix C.

Other monitoring methods

There are a range of other monitoring methods, that can be used to identify sources of exposure and assess control effectiveness, as outlined in sections 6.2, 6.3 and 6.4 of RS14.

7.4 Investigating exposure exceedances

If a single personal full shift exposure sample result, exceeds the relevant shift adjusted exposure limit, an investigation should be initiated. The investigation should determine the absent or failed control/s that contributed to the measured exposure, and identify the actions required (short and/or long term) to ensure welding fume exposure is as low as reasonably practicable. Reference should be made to the range of welding fume control options listed for common welding methods in Appendix E.

At a minimum, the investigation should consider the following:

- date of sample
- SEG
- work location
- crew
- activities/tasks carried out (including on-arc times, weld method, and other hot work tasks performed).
- Personal Protective Equipment (PPE) used and for what activities/tasks (including times).
- controls in place. Refer to control options listed in Appendix E.
- base metal consumables (e.g. mild steel, consider coatings)
- environmental wind conditions (still, light breeze, windy)
- environmental location (indoors, outdoors, confined space)
- operators head position (inside plume, outside plume)
- adjacent activities contributing to exposure (other welding tasks in close proximity)
- maintenance schedule vs actual maintenance records for all equipment (e.g. PAPR, on-tool, welding equipment, extraction)
- equipment pre-start checks/status of welding fume controls
- sampling data (determine if sample valid)
- operational conditions (e.g. normal/maintenance, shift/downtime).

The results of the investigation should be recorded and analysed in conjunction with previous investigation findings, to identify any trends and changes required with the coal mine's SHMS. Any subsequent changes made to the SHMS must also be recorded and communicated to affected CMWs.

7.5 Reporting

Communication of personal exposure results

At the end of each sampling period, personal exposure sample results should be communicated to individual CMWs that participated in the monitoring and de-identified results made available to all CMWs at the mine, regardless of the measured concentration.

Annual sampling result

Individual mines will be requested by RSHQ at the end of each calendar year, to forward the results of personal exposure sampling for welding fume. Sampling records including SEG results should be submitted to RSHQ in the approved format provided.

7.6 Assessment of exposure risk

Estimating SEG exposure

Assessment of exposure risk for individual SEGs should be conducted by a COH or equivalent.

For the purpose of this guidance, the exposure profile for each SEG should be estimated separately and compared with the OEL to determine if the exposure is acceptable, unacceptable or uncertain.

- Where monitoring data is available for the SEG, the exposure should be calculated using quantitative analysis. This includes welding SEGs identified for minimum sampling requirements (Table 4) and/or other groups monitored at the mine.
- Where monitoring data is not available, the exposure for this welding SEG should be estimated using semi quantitative analysis, by comparing with similar groups and activities that have been monitored and using professional judgement. This includes work groups not listed for minimum sampling requirements (Table 4). (EN 689 2019)

Following the collection of sufficient samples, inferential statistics should be generated to summarise the dataset and estimate the exposure profile for the SEG. Various tools can be used to generate descriptive statistics, as outlined in RS14. This includes the free industrial hygiene statistical excel tool from the American Industrial Hygiene Association (AIHA n.d.).

To assess and determine if risk is acceptable the analysis should calculate the average exposure concentration for the welder workgroup and SEG (6.1.2). Professional judgement may be required, due to inherent variability of welding fume exposure concentrations and should include consideration of the following:

- average (mean/Minimum Variance Unbiased Estimate (MVUE))
- measures of variability Geometric Standard Deviation (GSD)
- confidence intervals (95% upper confidence limit)
- range (min-max).

These will determine whether further sampling is required.

Where task based sample data is being analysed consideration should be given to welding activities performed during the shift and whether the sampling is representative of all welding fume exposure in the shift. Where task

based sampling is not representative, the analysis of this data can only be used for qualitative assessment. Additional methods to assess exposure may include:

- reasonable worst-case measurements (maximum exposures likely to occur with controls in place)
- estimating exposure (using appropriate models or algorithms)
- comparison with other similar operations and welding activities.

When qualitative methods for assessing exposure are inadequate to estimate exposure, additional sampling is recommended. (EN 689 2019)

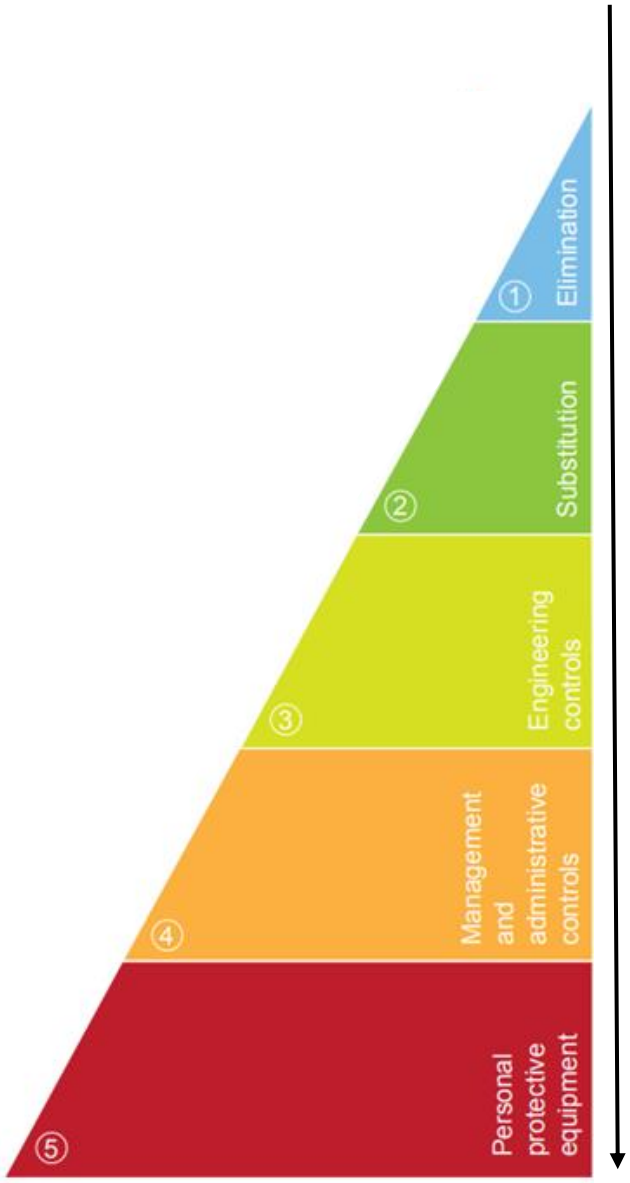
7.7 Monitoring program review

Site should review their welding fume monitoring program on an annual basis and submit their review to the SSE.

If there are significant changes to the plant, equipment, operating environment, operating methods, or SEGs that have the potential to impact welding fume exposure levels, or there is a significant change to the exposure estimate following periodic monitoring, the frequency of sampling must be reviewed.

8.0 Welding fume controls

8.1 Hierarchy of controls



Elimination will require the need to completely remove the hazard from the workplace by replacing it with alternatives – *There are limited examples of practical ways to eliminate the need for hot work and welding required in coal mining.*

Replacing/substituting either the base metal or the welding type/technique to reduce exposure.

Examples include substituting FCAW with MIG, SS FCAW and GS FCAW.

Engineering design to reduce exposure with the use of equipment, tools and design.

Examples include robotic welding, on tool extraction, local exhaust ventilation (fume capture hoods).

Administrative procedures, documents and training that assist in control effectiveness to reduce exposure.

Examples include adequate maintenance schedule and pre-checks, worker rotation, worker positioning and training packages.

Personal protective equipment to reduce exposure to welding fume.

Examples include the correct selection, use and maintenance of respiratory protection with focus on the use of PAPR.

Figure 10: Hierarchy of Control as applies to welding fume (British Occupational Hygiene Society n.d.)

To achieve an acceptable level of risk and minimise exposures to welding fume to as low as reasonably achievable, the selection of control measures to reduce welding fume, should follow the hierarchy of controls with higher order control measures considered first.

In practice the management of welding fume will require a combination of different order controls to be applied, to be effective.

A summary of the fume control options available for common welding tasks performed in Queensland coal mines has been tabulated in Appendix E.

8.2 Elimination

Alternative methods such as mechanical/chemical joining techniques, that eliminate the need for hot work and welding are not suitable for the types of repair work required in coal mining and are outside the scope of this guidance note. Should more information become available on suitable alternatives, this guidance note will be updated.

8.3 Substitution or modification of process

Substituting or modifying the welding process can reduce the welding fume emission rate and potential for exposure. A change management process should always be conducted to ensure the quality and standard of weld specified by the manufacturer of the plant being repaired, is not compromised. In some cases, the alternative process will have to be trialled and validated, for the manufacturer to authorise the change.

There are a large range of alternative options available for selection, to reduce the generation and concentration of airborne welding fume during welding.

8.3.1 Welding method

The emission rate of welding fume is primarily determined by the welding method used, as summarised in Figure 2. Where feasible, selecting lower fume-emitting welding processes can significantly reduce airborne fume concentrations. For example, opting for a GMAW process such as MIG with solid wire, instead of FCAW, or choosing gas-shielded FCAW over gasless FCAW for the same task, can substantially reduce the volume of welding fume generated.

However, these substitutions may require additional skill and tighter weld procedure control to maintain weld integrity. For instance, when replacing FCAW with MIG, weld quality can be maintained using waveform control and targeted training to improve the welder's technique.

Process-specific considerations include:

- FCAW remains appropriate for welding in the field or where the welding location cannot be effectively shielded from wind.
- Solid wire MIG is suitable for repairs and allows for faster welding, provided the operator uses spray transfer or pulsed mode. Note that the shielding gas composition, particularly those with high CO_2 content may limit the ability to run in these modes.
- Solid wire MIG in spray mode produces approximately one-third the fume of FCAW under comparable conditions.
- Caution should be exercised when attempting deep single-pass welds (e.g. 8 mm) with MIG, as this can lead to metallurgical issues such as centreline cracking. Mines may not be aware of these risks if weld procedures have not been appropriately qualified.

- New generation plasma gouging equipment can produce significantly less fume than traditional air arc gouging for certain applications, specifically those involving the removal of small volumes of weld material with relatively shallow penetration requirements.

8.3.2 Welding machine

The use of modern inverter-based welding power sources, especially synergic pulse power sources, is known to enhance arc stability compared with the use of older transformer-based power sources. Maximising the stability of the welding arc is essential to minimising excess weld spatter, and this is known to reduce welding fume. This can also be achieved by optimisation of weld procedures, and ensuring that the welding torch, cables and electrical connections are well maintained.

Waveform-control technologies commonly utilised in both pulsed spray mode and controlled short circuiting (dip) mode for the solid wire GMAW process for example, permits welding at lower average amperages and voltages than would be otherwise possible whilst maintaining good control and stability of the process, including productivity and arc penetration characteristics. Use of pulsed spray mode can reduce welding fume produced by up to 80%.

Waveform controls are not effective on the FCAW process other than where metal-cored wires are utilised.

8.3.3 Shielding gas

Shielding gas primarily comprised of a mixture of inert gas typically Argon (Ar) generates less welding fume compared with active gas which is predominantly comprised of Carbon Dioxide (CO_2).

Where possible, for gas shielded welding process, select a shielding gas with low CO_2 content. Replacing an Ar/18%CO_2 shielding gas mix with an Ar/10%CO_2 mix can reduce the volume of welding fume evolved by up to 50% (Emily Quecke 2023). This option is effective for both the GMAW and FCAW process (including metal cored wire). When substituting shielding gas, consideration should be given to the manufacturers' recommendations for specific consumables.

Consumable electrodes and wires classified for use with an Ar/10% CO_2 (Standards Australia 2022) shielding gas mix should be selected.

8.3.4 Electrode/wire composition

There are a range of consumable electrodes and wires with compositions designed to minimise the fume generation rate and toxicity of fume generated. For example:

- for the GMAW and FCAW processes, copper-free wire consumables are available, that have been confirmed to generate less fume than copper coated wires
- new low-Mn seamless FCAW wires are available to minimise Mn emissions
- chromium free wires are available as an alternative consumable wire for use in hard facing processes.

8.4 Isolation/separation

Remotely controlled and operated welding equipment removes the boilermaker from the location where the welding process is applied and from where the welding fume and gases are being generated.

8.4.1 Robotic welding

Robotic welding provides one option for remote controlled welding, allowing the workers to operate the equipment from a safe distance or even from separate control rooms. This not only reduces the risk of inhalation of welding fume and gases but also limits exposure to other welding-related hazards, such as intense UV radiation and heat. Robotic welding systems can be equipped with a variety of welding processes, including MIG and TIG to accommodate different materials and joint types.

There are two primary categories of robots:

- Cobots, or collaborative robots, are designed to work alongside human operators in a shared workspace and can also be operated remotely. They are typically smaller, more flexible, and equipped with advanced sensors and safety features that allow them to detect and avoid harmful contact with humans. Cobots are particularly useful in applications requiring frequent adjustments or a high degree of human-robot interaction. The smaller portable size enables them to be relocated to weld applications in situ, which is an advantage for large mining equipment shutdowns including draglines.
- Industrial robots, are larger, more powerful machines designed for high-speed, high-volume production environments. They are ideal for repetitive, high-volume welding operations. The larger fixed application of these robots has limited application on coal mine sites, where large welding jobs are typically performed outside the workshop and in the field.

Robotic welders bring significant advantages, including improved weld quality, reduced material waste, and enhanced worker safety. The increased weld capacity has the potential to introduce additional exposure hazards which require additional controls to be implemented. For example, the increased weld rate/temperature may generate higher welding fume concentration and the additional weld material deposited, requires an increase in grinding performed by the boilermaker each shift.

There is one coal mine site in Queensland that has commenced trialling the use of cobot welding for dragline shutdowns. This trial has successfully demonstrated the cobots effectiveness at completing welding activities, particularly in applications and positions which are less suitable for humans. These include welding over prolonged durations in awkward positions and inside confined spaces.

Figure 11 and Figure 12 show the trial cobot welder setup in various locations on a dragline shutdown.



Figure 11: Shows cobot with boilermaker performing weld repairs on a dragline.

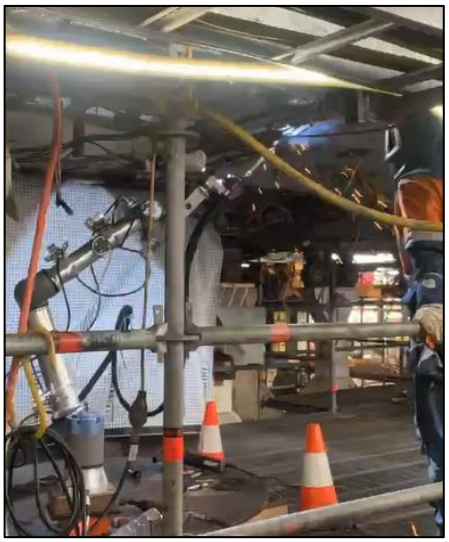


Figure 12: Shows cobot with boilermaker performing weld repairs on a dragline.

8.4.2 Submerged arc welding

Submerged arc welding is a process that creates an arc under a bed of flux thereby containing the evolution and emission of welding fume into the surrounding air. It is mainly used in the manufacture of large components of mining equipment, before reaching the mine site. There is limited application of submerged arc welding on the mine site where hot work is primarily directed at repair type welding, often in situ.

8.5 Engineering – ventilation

8.5.1 Types of ventilation

Static ventilation

In static wind conditions welding fume tends to rise upwards and disperse due to its thermal buoyancy. It gradually cools and reaches thermal equilibrium with the surrounding air, after which it falls back down. Maximum fume concentration occurs at the weld source and often between the weld source and the boilermakers head during most conventional manual welding practices. The concentration naturally decreases with distance away from the weld as the fume rises and disperses.

Natural ventilation

Natural ventilation in the mining context typically refers to the wind conditions which are present in the absence of any mechanical ventilation system. Natural ventilation occurs in the outdoor environment, or as flow through ventilation passing through buildings and workshops which are opened to the external wind conditions, through roller doors, windows and louvers. To a lesser extent natural ventilation can occur inside buildings as the result of temperature differences causing convection flow resulting in an up draft toward the roof (AS/NZS 4740 2000). Roof extraction fans may need to be used to supplement the natural air flows generated by convection currents.

In general, natural ventilation is unreliable and not effective at reducing the fume concentration at source or in the area around the boilermaker's head and breathing zone, during conventional welding practices. At sufficient

wind speeds, natural ventilation can provide sufficient air flow to displace and blow the fume away from the boilermaker's head and breathing zone, if the boilermaker is positioned upstream of the wind direction. Most of the welding in coal mining is performed outside in natural ventilation.

To minimise exposure to welding fume in natural ventilation conditions the boilermaker must deliberately position themselves and their head away from the fume cloud.

General ventilation (mechanical ventilation)

General ventilation is also known as mechanical ventilation in mining and refers to the supply and exhaust of air from an area or room, using mechanical fan ventilation systems. Fans are used to control the direction, quantity and rate of air flow to a designated location. Fans should not be positioned to direct welding fume towards other workers. Welding screens can be used to assist with controlling the direction of air flow.

Where possible, separate welding bays and/or designated welding workshops should be established and segregated from other workshop areas to prevent welding fume contamination and secondary exposure risks for other CMWs.

Dilution ventilation

Dilution ventilation occurs where positive air flow containing uncontaminated air is introduced to the plume to further disperse and reduce the fume or gas concentration.

Exhaust ventilation

Exhaust ventilation is where negative pressure air flow is introduced to capture and remove the airborne contaminants. There are a range of different exhaust ventilation systems available to control welding fume and suited to different applications. Each type has advantages and disadvantages depending on the practical application. The closer the extraction point is to the weld the more effective at controlling the fume generated (Figure 13).

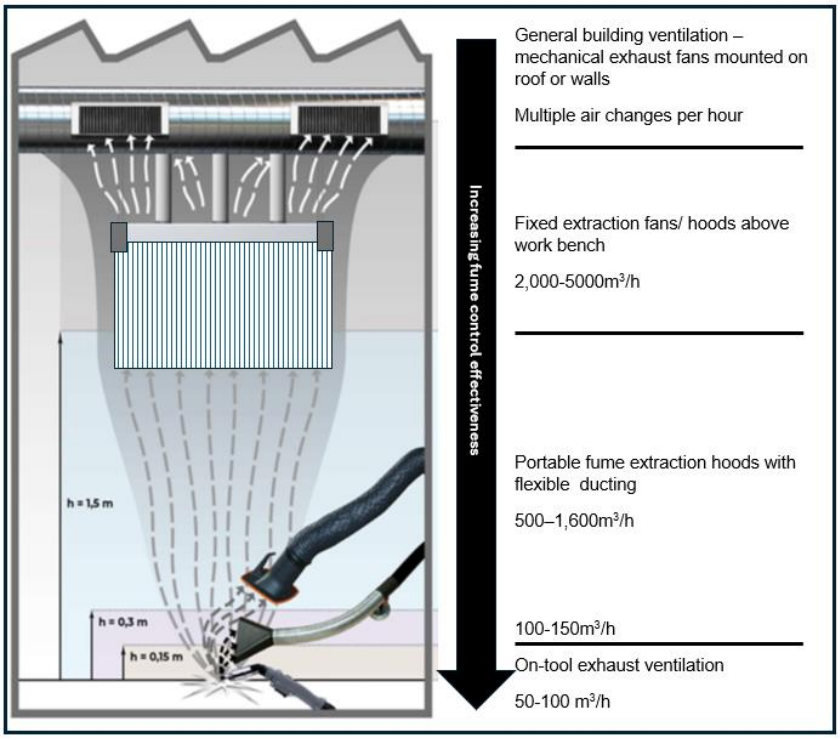


Figure 13: Exhaust ventilation and the proximity to the weld source results in more effective fume capture (European Welding Association 2024).

Local exhaust ventilation (LEV)

Local Exhaust Ventilation is where the extraction is positioned to capture the fume at source before it reaches the breathing zone. The two types of exhaust ventilation commonly used in practice and considered in this guidance note are the on-tool (on-torch) and welding fume hood.

The use of other ventilation systems for welding fume control including fixed welding booths, welding bench enclosures/hoods, and general/displacement ventilation exhaust fans applied to buildings, are not considered practical for the typical welding applications in coal mining, which are generally outdoors and conducted in situ on large fixed or mobile plant.

Capture velocity is the minimum air flow rate required by the exhaust ventilation to capture and remove the welding fume and is typically 0.5-1m/s.

These fume extraction systems are typically (not always) fitted with particulate filtration media to capture the collected welding fume before it is discharge to atmosphere. The filter media should be rated/labelled as High Efficiency Particulate Air (HEPA) class filtration. The fixed systems typically have higher capacity filtration systems and require less frequent replacement compared with the smaller portable units. Consideration should be given to controlling the potential risk of secondary exposure during cleaning and replacement of filtration systems.

8.5.2 On-tool exhaust (form of LEV)

The modern on-tool exhaust system has fully integrated local exhaust ventilation into the hand piece and has the capture vents positioned to provide a suitable capture velocity where the fume is generated, while minimising the risk of removing shield gas and compromising weld integrity. The exhaust ventilation is typically provided by a vacuum system which is separate to the weld machine/shield gas and is either a stand-alone portable system or can be plumbed into an existing fixed extraction system.

The performance and effectiveness of on-tool fume exhaust systems depends on a number of factors including technique, positioning, orientation (angle) and direction. For example, fume exhaust works best using down hand position and is least effective when vertical or overhead. Numerous studies have been conducted to estimate the effectiveness of fume reduction from using on-tool exhaust systems (Knott, et al. 2023). It is estimated the fume reduction will range between 50 and 95 percent (Knott, et al. 2023).

There are a range of on-tool exhaust systems commercially available in Australia for most FCAW, MIG and limited TIG applications. These include fixed or portable systems. A comparison of modern on-tool hand pieces and traditional units are shown in Figure 14 and Figure 15. The weight of on-tool exhaust handpieces are heavier than conventional units, due to the extra exhaust tubing. They can be up to 0.5 kilograms heavier compared with conventional units. This accounts for the trailing cables/exhaust tubing and is measured at a height of 1.5 metres above the ground.

There are a number of coal mines where boilermakers are using on-tool exhaust for routine welding jobs, including both fixed systems inside the workshop and portable systems outside in the field (Figure 16 and Figure 17).



Figure 14: Close up of on-tool exhaust unit.



Figure 15: Comparison of on-tool exhaust unit (bottom) vs traditional unit (top).



Figure 16: Fixed on-tool exhaust unit set up inside workshop.



Figure 17: Portable on-tool exhaust unit set up inside workshop.

On-tool Exhaust	
Benefits	Limitations
Up to 90% fume capture (down hand). Captures fume at source of generation before reaching the breathing zone. Capture efficiency less susceptible to cross wind interference. Integrated/attached to weld torch hand piece. Activated automatically with weld trigger. Portable units can be used on large structures and outside in the field. Extraction system can be used as secondary vacuum clean up. Extraction system follows the source, isn't required to be manually positioned.	Capture efficiency depends on weld position and technique/angle e.g. significantly reduced for overhead and vertical. Application restricted to FCAW, MIG and TIG equipment. Not suitable for stick, plasma, or arc air gouging. Not effective for capturing O3 gas generated away from the arc. Additional HEPA filtration required to capture contaminants in exhaust discharge. These filters require regular replacement and handling that can lead to secondary exposure risks. On-tool exhaust systems are heavier by up to 0.5 kg compared with conventional setups and this may increase ergonomic risks when welding out of position. The shield gas can be disrupted when the weld tip/shield gas shroud is inserted into small spaces with limited volume and this requires manual adjustment of the exhaust flow rate.

8.5.3 Fume extraction hoods

Fume extraction hoods are completely independent of the weld equipment and are designed with adequate flow rate to provide sufficient capture velocity to entrain and capture the fume away from the boilermakers breathing zone.

The effective capture distance out from the extraction hood is specified by the manufacturer for each device and for a standard 150 millimetres and 200 millimetres diameter hood this is typically 300 millimetres and 450 millimetres, respectively. To be effective the distance between the hood and the weld must be within the capture distance of the hood, and in practice this is approximately one to two diameters. The capture distance is also influenced by the positioning of the extraction hood relative to the weld. For standard hoods this is typically positioned vertically and directly above the weld in the direction of thermal up lift, or on the bench as shown in Figure 18. Placing the hood horizontal and in contact with the bench has the effect of extending the hood. Some modern designs include a flat edge on one side of the hood to facilitate placement onto the bench top for this purpose. Placing the hood at 45 degrees to the weld is the least effective position (Figure 19).

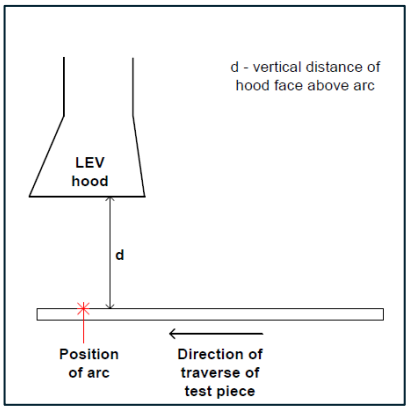


Figure 18: Hood positioned directly above the weld.

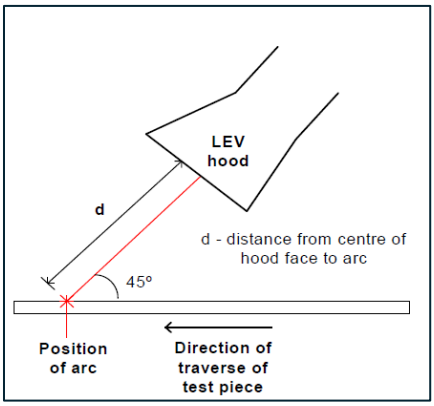


Figure 19: Hood positioned at 45 degree angle to the weld.

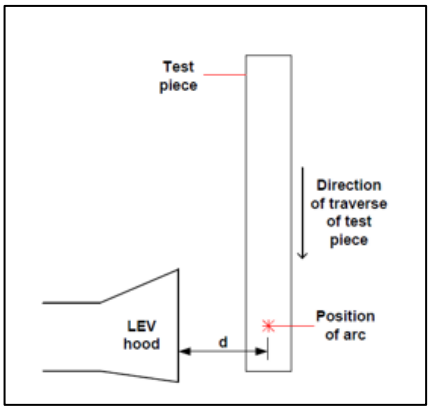


Figure 20: Hood positioned adjacent to weld piece.

The performance and effectiveness of fume extraction hoods primarily depends on the positioning of the hood in relation to the weld and this requires the boilermaker to continuously adjust and maintain optimal hood positioning while performing the weld. In practice the optimal hood position is typically not able to be maintained consistently for situations where the boilermaker moves location.

The capture efficiency from fume hoods is also susceptible to cross wind which interferes with the capture velocity. It is estimated from laboratory studies that the fume reduction when hoods are in the optimal position with no cross draught and welding in one location, will be up to 80 percent. Simulated in-field testing in workshop conditions has indicated the fume capture efficiency was approximately 90 percent (Knott, et al. 2023). Optimal performance is rarely achieved in practice in coal mining where the boilermaker does not weld in the one location and there are always cross drafts present.

Fume extraction hoods are commercially available in a large range of fixed and portable configurations and can include single or multiple hoods connected to the same system.

Fume hoods are not limited to the type of weld method or consumable being used as is the on-tool exhaust system and hoods can be utilised in a large range of indoor and outdoor scenarios.



Figure 21: Fume extraction hood set up on bench.



Figure 22: Fume extraction hood during grinding activities (30).



Figure 23: Fume extraction hood set up between the weld and the boilermaker (30).

Fume extraction hoods	
Benefits	Limitations
Up to 80% fume capture (when positioned correctly). Draws fume away from breathing zone for capture. Variety of fixed and portable configurations available. Captures O3 gas generated in proximity to arc. Easy to apply once installed.	Limited capture distance from hood and this is defined by hood design and flow rate. Capture performance is affected by cross wind interference. Capture performance highly dependent on positioning of hood (distance from weld and orientation). Requires frequent manual positioning for weld length greater than approximately 100-150 millimetres. Applications generally limited to welding inside workshop and at fixed locations. Additional HEPA filtration required to capture contaminants in exhaust discharge. These filters require regular replacement and handling that can lead to secondary exposure risks.

8.5.4 Pedestal fan



Figure 24: Pedestal Fan are commonly used in workshop adjacent to welding activities.

Pedestal fan	
Benefits	Limitations
Fans readily available, convenient to set up and low cost.	Does not capture or remove fume.
Fan can be positioned distant from the weld and not interfere with activity.	Disperses fume to other work areas nearby and may increase exposure to other workers.
Less susceptible to cross wind interference.	Effectiveness dependant on positioning of operator relative to weld source and fan direction.
Effectiveness less susceptible to distance from source (meters).	Air flow can disrupt shield gas and compromise weld integrity.
Air flow contributes to body cooling of operator.	

8.5.5 Mechanical exhaust fans

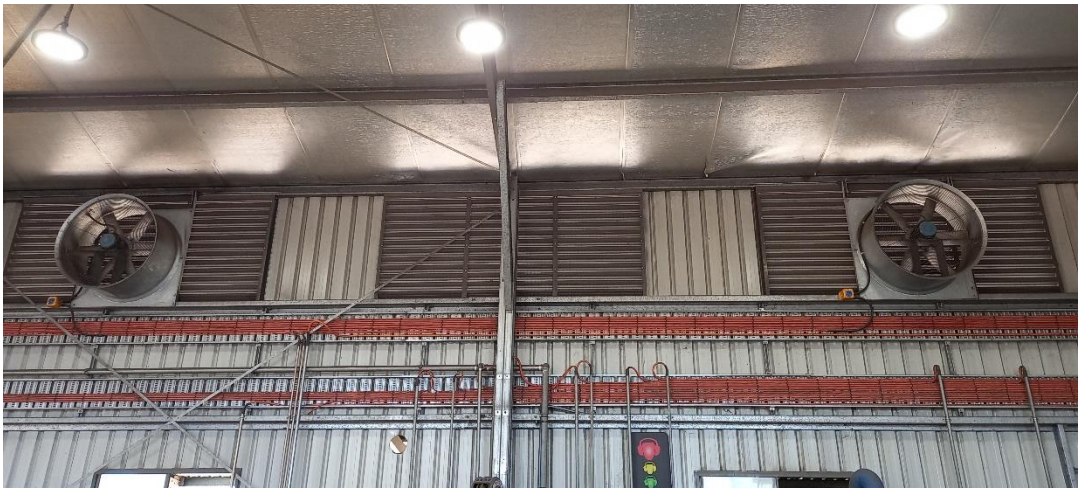


Figure 25: Wall mounted exhaust fans positioned in ceiling space.

Mechanical exhaust fans	
Benefits	Limitations
Fans readily available, convenient and low cost.	Does not capture or remove fume at source.
Removes fumes from roof space and reduces accumulation inside building.	Does not reduce fume exposure for boilermaker.
Reduces exposure to other workers inside building.	Disperses unfiltered fume to outside atmosphere and may increase exposure to others outside building.
Air flow contributes to general building ventilation and body cooling of operator.	Fan cannot be easily re-positioned or moved.
	Fans may increase noise levels inside the workshop.

Weld integrity

Air flow passing over the weld may interfere with shielding mechanisms and potentially compromise weld integrity. Consequently, there is a tendency for boilermakers to prefer areas of low air movement when setting up to weld, and there is a reluctancy to introduce additional air movement close to the point of welding. Testing conducted on welding fume extraction hoods and on-tool exhaust systems have demonstrated that weld integrity should not be adversely impacted by introducing controlled ventilation at flow rates up to 0.5 m/s for most conventional gas-shielded arc welding methods.

When welding is conducted with the weld tip and gas shield shroud positioned inside small cavities with limited volumes of air, the flow rate on LEV systems may need to be adjusted to prevent the removal of shielding gas. For example, some on-tool systems have a local bypass valve to enable the boilermaker to manually restrict the extraction ventilation flow during welding in small spaces. Some also have flow meters to test and calibrate the exhaust flow rate and the shield gas supply flow rate, to ensure optimal performance.

8.5.6 Maintenance of ventilation controls

The performance and effectiveness of ventilation systems depends on how well they are maintained, and this requires regular periodic inspections and servicing.

A daily prestart operational check should be conducted prior to use of the ventilation control. For LEV this should include the following:

- Visual inspection of inlet, hood, hosing/ducting for visible damage
- Visual inspection of condition of filter media and change where required.
- Turn on LEV and check air is drawing across inlet/hood.

Exhaust ventilation systems should be designed and equipped with the required pressure gauges, flow meters, and/or alarm indicators, to enable ongoing performance checks in-service and periodic maintenance inspections. This is required for ongoing control effectiveness verification.

Ventilation control equipment should be incorporated into the mine's scheduled preventive maintenance program, and work order system, requiring inspections at least every 12 months (AIOH n.d.). The results of inspection and measurement should be recorded and rescheduled within specified intervals. The key maintenance inspection requirements for on-tool exhaust systems and fume exhaust hoods have been listed below.

An example of a scheduled maintenance inspection checklist for exhaust ventilation systems has been provided in Appendix G.

On-tool exhaust systems

- Visual inspection of inlet nozzle for obstructions, vacuum hose for holes and/or deformation.
- Visual inspection of filter media and housing for signs of breakthrough, by-pass and/or build-up of surface deposition and verification of filter replacement frequency.
- The volume flow rate at the extraction nozzle(s) should be measured and compared with exhaust specification.
- Back pressure drop across the filter media should be measured and compared with specification.

- Shield gas supply flow rate should be measured and checked against specification.

Fume exhaust hoods

- Visual inspection of hood and face inlet for obstructions and damage.
- Visual inspection of flexible and rigid ducting for damage, collapse and/or penetration holes.
- Visual inspection of filter media and housing for signs of breakthrough, by-pass and/or build-up of surface deposition and verification of filter replacement frequency.
- Face velocity across the hood face inlet surface area should be measured and compared with specification, for example using a hot-wire anemometer
- Back pressure drop across the filter media should be measured and compared with specification.

Where faults are found on inspection the ventilation control equipment should be removed from service and repaired. Interim controls should be implemented until the device is back in service.

8.6 Administrative Controls

Task Rotation

For prolonged high-exposure tasks, rotation should be considered to minimise individual exposure duration. However, rotation may be limited by the availability of workers with appropriate qualifications and skills to perform competent welding. In many cases, the most experienced or efficient worker performs the majority of the tasks. Where feasible, rotation should be planned and documented.

Worker Positioning

Positioning of the welder's head away from the fume can reduce potential exposure. For example, set up and position the welder's head and breathing zone upwind of the fume's direction of flow or move work pieces to the side where possible. Further information on positioning is provided in Fume Minimisation Guidelines (Appendix J)

Welding Methods and Procedures

The site's SHMS should include safe operating procedures (SOPs) for managing hazards associated with hot work activities including welding. As a minimum, these should incorporate the following.

For hot work and welding procedures:

- identify welding fume and gases as key hazards
- specify what types of ventilation controls will be applied to each task (e.g., on-torch extraction, fume hoods, ducted exhaust systems)
- what administrative controls will be applied (e.g., welder positioning and job scheduling)
- list of the minimum respiratory protection requirements (including filtration class) for hot work and welding activities
- specify what fume controls will be applied for welding in confined spaces (e.g. ventilation plans Appendix H)

- outline the training requirements for hot work, including induction and refresher frequency
- consider fume dispersion mapping, to assess the potential exposure to trade assistants or other workers positioned adjacent to the welder. This supports more informed decisions around task positioning and timing.

Hot work permits should specify:

- the type(s) of hot work activity being undertaken
- the types of ventilation controls to be put in place
- the type and level of respiratory protection to be used.

Training and Competency

Training packages on welding fume hazards, control measures, and correct PPE use, should be developed and delivered to all boilermakers performing hot work activities at the mine. Refer to Section 9 below for further details.

8.7 Respiratory Protection Equipment (RPE)

RPE is the least effective form of hazard control on the hierarchy of controls. The objective should be to, wherever possible, reduce exposures to below the OEL and to as low as reasonably achievable, without considering the potential protection afforded by RPE. Every effort should be made to control exposures by higher order controls before considering the use of RPE.

However, currently, where welding and other hot work processes are conducted on mine sites, the concentration of fume and gas generated will require the wearing of RPE in addition to engineering and other higher order controls, to protect the boilermaker from exposures above recommended exposure limits.

The mine is required to have a SOP outlining how to use PPE for tasks at the mine and this is to include training on selecting and using RPE. The PPE SOP should specify the type of respiratory protection required to be worn for different types of hot work activities including welding at the mine, and other minimum requirements including facial fit testing and training.

There are a range of suitable respiratory protection types available for protecting the boilermaker from exposure to welding fume. These range from negative pressure disposable and reusable half face masks to PAPR and air-supplied head coverings/helmets connected to a compressed air supply.

There are a broad range of different product manufacturers for each type of respirator and examples of commonly used RPE types are detailed in the RSHQ fact sheet on RPE and Respiratory protective equipment at work (Health and Safety Executive 2013).



Figure 26: Example of commonly used disposable half face RPE

Disposable half face masks suitable for welding fume include P2 or P3 particulate filters and are impregnated with carbon/charcoal to remove nuisance gases. These are negative pressure RPE, and quantitative facial fit testing is required for the individual wearer to ensure correct size and shape is selected for the individual.



Figure 27: Example of commonly used reusable half face RPE

Reusable half face masks provide a higher level of facial fit/seal and enable higher capacity filter cartridges to be used. These facepieces should be fitted with filter cartridges rated for welding protection. As a minimum this should include P2 or P3 Class filters. Organic A Class filters, Aus Class or nuisance/ odour organic filters may be used for certain applications. Gas filters should always be fitted in addition to and in combination with particulate (P2 or P3) filters and never in isolation. Particulate filters are always required to remove welding fume. These are negative pressure RPE and quantitative facial fit testing should be conducted on individual wearers to ensure corrected size and shape RPE is selected for the individual.



Figure 28: Example of commonly used PAPR

Powered air purifying respirators (PAPR) are typically incorporated into a welding style helmet/face shield, where the fan and filter assembly are mounted onto a belt and the filtered air is supplied through a hose connected into the back of the helmet. The filtered air is supplied into the helmet at a specified minimum flow rate (usually 170 lpm) in order to maintain positive pressure inside the loose fitting head covering/helmet. The filter housing is capable of fitting particulate P2 or P3 filter cartridges alone or in combination with nuisance/ odour or A Class organic gas filter cartridges.

When well maintained, PAPRs are highly effective in providing protection for the boilermaker. Their use is not recommended in confined spaces where air-supplied systems are preferred.

Limitations: The performance of PAPR welding helmets depends on the integrity of each of the many components and this requires regular maintenance inspections. In practice, the protection provided by PAPR is typically compromised when:

- there are leaks or obstructions in the hose connecting and delivering filtered air to the helmet
- the flow rate delivered into the helmet falls below the minimum required to maintain positive pressure
- facial seals are loose and allow inward leakage into the breathing zone.

Air supplied or air-line respirators typically use the same loose fitting welding style helmet/face shield used for PAPR, except the belt mounted fan and filtration assembly is replaced with a low pressure air-line for connection back to a regulated compressor outlet. An in-line filter assembly including particulate, water vapour and oil trap, is installed between the compressor and respirator to prevent contaminants from the compressor entering the respirator. The air-line configuration should be regularly tested for conformance with the minimum air quality requirements specified in AS 1715:2009 (Standards Australia 2009). Testing is typically performed at 6 or 12 month intervals depending on the intensity of use.

Table 6: Selection, fit testing, and planned inspection of RPE* for hot work activities

RPE Type	Filter Class	Fit Test Required	Planned Inspection Schedule
Disposable	P2	Annually	N/A – discard after use
Reusable half face	P2/P3	Annually	Daily pre-start inspection
PAPR	P2/P3, ABE	N/A	3-6 monthly
Air supply air-line	Inline filter	N/A	3-6 monthly

* Note: In addition to RPE types listed above, further consideration should be given to the ventilation conditions present, when selecting RPE.

Maintenance/Inspection Checklist

Daily pre-start inspection:

Disposable masks (e.g. P2):

- conduct a fit check every time the mask is worn
- discard after each use
- replace multiple times per shift if required (e.g. visibly soiled, wet, or damaged).

Reusable masks **(Half/Full Face/PAPR):**

- inspect before each use for damage
- check inhalation/exhalation valves (where appropriate)
- conduct a fit check every time the mask is worn
- clean after each use (daily or weekly depending on use frequency)
- ensure battery fully charged (where appropriate)
- store in a clean, dry, sealed container when not in use
- date-stamp and track filter cartridge replacements.

Filter Replacement Guidelines:

- Particulate Filters (P2/P3): Replace when visibly soiled and/or breathing resistance increases (typically weekly or monthly depending on weld intensity).
- Gas Cartridges: Replace within 6 months after opening sealed packaging (even if not used), and after approximately 1-4 weeks of use, depending on weld intensity, or sooner if breakthrough smell or odour is detected.
- Filter cartridges should not be used after the expiry date stated by the manufacturer/supplier, on the filter packaging.

3–6 Monthly Planned Maintenance

PAPR Assemblies:

- add to site's planned equipment maintenance register for inspection every 3–6 months as a work order
- inspect entire unit (hood, hose, filters, battery) for function and wear
- test airflow and battery performance
- document maintenance actions.

An example of a planned 3-6 monthly maintenance inspection checklist has been attached in Appendix F: PAPR checklist.

All CMWs required to wear respiratory protection should be trained in the selection, use, maintenance and limitations of these devices. As a minimum this should include:

- donning and doffing the facepiece
- visually inspecting for faults and damage
- conducting a fit check
- how and when to replace filters (including both particulate and gas filter types)
- clean shaven requirements for close fitting facepieces
- completing an individual facial fit test to select the correct size and type of respirator.

8.8 Confined space

Welding in confined spaces within coal mining environments presents significant risks due to the potential accumulation of toxic gases and fumes, heat stress, and reduced oxygen levels, and should therefore be considered in the hazard identification phase. Each task must be assessed and managed on a case-by-case basis, as conditions can vary widely depending on the location, materials, and ventilation available.

It is essential where hot work/welding activities are to be performed, that welding fume is clearly identified as a hazard and acknowledged on the confined space entry permit, and that a site-specific ventilation plan is developed and implemented to ensure the safe dilution and extraction of fumes and gases.

An example of ventilation plan drawings applied to shutdown welding work conducted inside a dragline, is provided in Appendix H.

The specific types of welding methods and ventilation controls being applied to control welding fume inside the confined space should be clearly identified and documented in the risk assessment, hot work permit, and confined space entry permit. The types of respiratory protection and filtration media required to be worn inside the confined space to protect CMWs from welding fume exposure should also be clearly documented in these permits.

9.0 HEALTH SURVEILLANCE

9.1 Conducting health surveillance for welding fume

The SSE and employers have obligations for ensuring the health surveillance of coal mine workers identified as being likely to be exposed to welding fume. The purpose of health surveillance is to:

- monitor for changes to a worker's health from baseline at the earliest opportunity
- implement early intervention measures to protect workers' health
- ensure that exposure controls and monitoring programs are effective.

Health surveillance shall be carried out by, or under the supervision of, an appointed medical adviser (AMA) under the CMWs' Health Scheme (the Scheme). The Scheme has been optimised to detect occupational lung disease early and ensures health examinations are performed to the required standards by appropriately qualified and experienced medical providers approved and regulated by RSHQ. It also ensures medical records are stored and maintained in an appropriate form, enabling comparative assessment to previous results.

Employers must undertake particular steps in the health assessment process in ResHealth[2] to ensure welding fume health surveillance is completed for each worker likely to be exposed to welding fume. Employers must ensure that exposure to welding fume is one of the hazards they identify in the employer section of the assessment. For workers identified as at risk of exposure to welding fume, health surveillance should be completed:

- before a worker commences work that may involve exposure to welding fume
- at frequencies specified under the Scheme, or as advised by the AMA.

and with consideration of the following:

- any personal exposure monitoring results
- specific welding fume tasks/exposures
- duration, frequency and level of exposure for workers
- effectiveness of engineering controls
- respiratory protection use (type and frequency of use).

As part of completing health surveillance, the AMAs may request and consider personal exposure monitoring results for welding fume, from the mine for relevant SEGs. The SSE and employer should facilitate the reporting of welding fume exposure monitoring results where requested by the AMA, to support the health surveillance process.

[2] Online system for the health assessment process under the Coal Mine Workers' Health Scheme

10.0 Education and Training

10.1 Education on Welding Fume

The mine's SHMS should document the training needs analysis for all CMWs involved in welding activities, to include education on welding fume health impacts and controls relevant to their mine. This should include initial training at induction to the site and job, as well as ongoing periodic refresher training.

Those CMWs requiring education on welding fume and controls should include:

- persons authorised to carry out welding activities
- supervisors of welding activities
- other CMWs who assist with welding activities, or who may be potentially exposed to welding fume. i.e. trade assistants, confined space stand-by person.

This education should include:

- the health effects of exposure to welding fume and the fume's various components
- how to minimise welding fume in the boilermakers breathing zone, including where to stand/operator positioning
- hot work SOP and hot work permit requirements
- what pre-use checks you should make to ensure your welding equipment is working correctly.
- which tasks have a high risk of exposure at the mine (confined space, bucket repairs)
- how to use controls, including
    - how to check and maintain the performance of welding fume controls
    - how to position movable LEV to make sure it is in the right place as you work
    - what to do if a control fails (PAPR, ventilation).
- how to select, use and look after RPE and PPE
- the safety risks associated with welding activities
- exposure monitoring for welding fume in the workplace, including:
    - exposure limits for each potential component
    - exceedance investigation process.
- health surveillance included in coal mine workers' health scheme.

In addition to the education outlined above, specific training and up-skilling of boilermakers may be required and should be considered, prior to introducing new fume control methods. This is to ensure the new control is applied effectively and any additional hazards introduced as a result of the change, are acknowledged and managed. For example, when substituting FCAW with MIG methods and/or using on-tool exhaust, for the first time.

Structured on-line courses are also available from a number of registered training providers and cover a range of topics from general awareness about welding hazards and best practice, to specific qualifications for welders and welding supervisors. For example, Weld Australia provides courses aligned with AS 1796 and AS2214.1, that are listed and accessible on their website.

Welding hazard awareness courses should be incorporated into the training needs analysis for welding supervisors, and for engineers and managers who oversee projects involving welding activities.

The site senior executive must ensure that welding activities are supervised to the extent necessary to ensure the appropriate level of competency as outlined under CMSHR s 85B Supervising workers.

11.0 Audit and Review

11.1 Review

The site's SHMS should provide for the regular review and continuous improvement of provisions relating to welding fume management to ensure continued suitability, adequacy and effectiveness. The management review process shall ensure that the necessary information is collected to allow the mine to carry out this evaluation. This should include performance and maintenance records for welding fume controls and sufficient welding fume exposure data for similar exposure groups undertaking hot work activities. This review shall be documented.

11.2 Audit

The site's welding fume management system should be subjected to regular audits to maintain relevance and ensure continuous improvement. The aim of the audit is to ensure that sections of the SHMS that relate to the monitoring and control of welding fume are effectively implemented and maintained. The audit should cover any associated risk assessments, hazard management plans, operating procedures, maintenance procedures (e.g. LEV and RPE), training records and personal exposure monitoring programs implemented at the mine.

The audit should also include the implementation of critical controls applied to welding fume, where these have been identified within the mine's PHMP for 'airborne dust and any other airborne contaminants'

Where the maintenance program is found to be ineffective in managing the operating performance of critical control systems actions are to be captured and documented to ensure the required changes have been implemented to rectify the inefficiency.

The frequency of the audit program should be based on the risk of the hazard concerned and the results of previous audits. The mine should establish and maintain procedures for the identification, maintenance, and storage of records.

Appendix A: Definitions and abbreviations

Term/Abbreviation	Definition
Brazing	Brazing includes flame and arc brazing (including “silver soldering”) where the melting temperature of the filler metal is above 450°C but always less than the melting temperature of the parent material.
Soldering	Soldering is a joining process where the melting temperature of the filler metal is less than 450°C,
Mild Steel	Mild steel, also referred to as *low-carbon steel*, is an iron alloy containing a relatively low carbon content—typically between 0.05% and 0.25% by weight. It also commonly contains manganese (0.3–0.9%) and silicon (~0.2%) and may contain trace amounts of other elements. Mild steel is characterised by moderate strength, high ductility, and excellent weldability, making it the most widely used form of steel across industrial applications. Its popularity is driven by its versatility, workability, and low cost. Various national and international specifications may fall under the general classification of mild steel—for example, AS/NZS 3678 Grade 250.

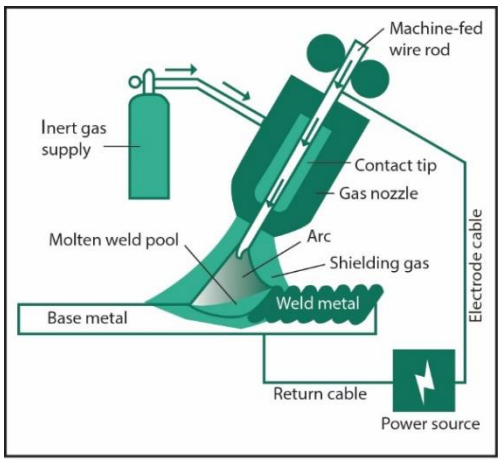
Term/Abbreviation	Definition
ALOR	Acceptable level of risk
CHPP	Coal handling and preparation plants
CMSHA	Coal Mining Safety and Health Act
CMSHR	Coal Mining Safety and Health Regulation
CMW	Coal mine worker
CO	Carbon monoxide
CO_2	Carbon dioxide
COPD	Chronic obstructive pulmonary disease
FCAW	Flux cored arc welding
FCAW-G	Gas-shielded flux-cored Arc Welding
FCAW-S	Self-shielded flux-cored Arc Welding
GMAW	Gas metal arc welding – often also termed MIG/MAG
GTAW (TIG)	Gas tungsten arc welding
HRA	Health risk assessment
ICMM	International Council on Mining and Metals
ISO	International Organization for Standardization

Term/Abbreviation	Definition
MIG	Metal inert gas
MMAW (Stick)	Manual metal arc welding – often used interchangeably with shielding metal arc welding or stick
MVUE	Minimum Variance Unbiased Estimate - the estimated arithmetic mean (average) of a lognormal distribution
NOx	Oxides of nitrogen
O_2	Oxygen
O3	Ozone
OEL	Occupational exposure limit
PAPR	Powered air purifying respirator
PE	Polyethylene
PHMP	Principal hazard management plan
PP	Polypropylene
PVC	Polyvinyl chloride
RPE	Respiratory protection equipment
ROM	Run of mine
RSHQ	Resources Safety and Health Queensland
SEG	Similar exposure group
SHMS	Safety health management system
SMAW	Shielded metal arc welding
SOP	Standard operating procedures
SSE	Site senior executive
SWA	Safe Work Australia
VOC	Volatile organic compounds
WRAC	Workplace Risk Assessment and Control

Ag	Silver	Mn	Manganese
Al	Aluminium	Mo	Molybdenum
Ba	Barium	Ni	Nickel
Co	Cobalt	O3	Ozone
Cr	Chromium	Pb	Lead
Cr(VI)	Hexavalent Chromium	Si	Silicon
Cu	Copper	Ti	Titanium
F	Fluorine	V	Vanadium
Fe	Iron	W	Tungsten
Mg	Magnesium	Zn	Zinc

Appendix B: Welding fume composition/common welding types

Gas Metal Arc Welding (GMAW)



GMAW, commonly referred to as "*MIG",* uses a continuous solid wire electrode, which melts due to the heat of an electric arc shielded by either an inert gas (e.g. argon or helium) or an active gas (e.g. Carbon Dioxide (CO_2) or an argon/CO_2/Oxygen (O_2) mixed gas). Fumes are generated mainly from the vaporisation of the electrode materials, the shielding gases used to protect the weld pool, and to a limited extent, the parent metal. The specific fume composition depends on the metals involved, such as carbon and carbon-manganese steel (e.g. mild steel), low alloy steel, or stainless steel. When working with mild steel, the fume cloud mostly contains a mixture of iron oxide, and manganese, and occasionally nickel. The fume from welding steels with low alloy consumables may also contain nickel and chromium.

Figure 29: Gas Metal Arc Welding (GMAW) or Metal Inert Gas Welding (MIG)

Various parameters can affect the types and quantities of gases generated. For example, all arc welding processes will generate O_3 and NOx due to the effects of UV radiation on the ambient air, especially at higher amperages. For shielding gas containing CO_2, carbon monoxide (CO) can also be formed within the arc.

NOTE: In Australia, the term *MIG* is commonly used to describe this process. ISO uses the terms MIG to refer to the process that uses an inert shielding gas and Metal Active Gas (MAG) is used to refer to the active shielding gas variant.

Flux-Cored Arc Welding (FCAW)

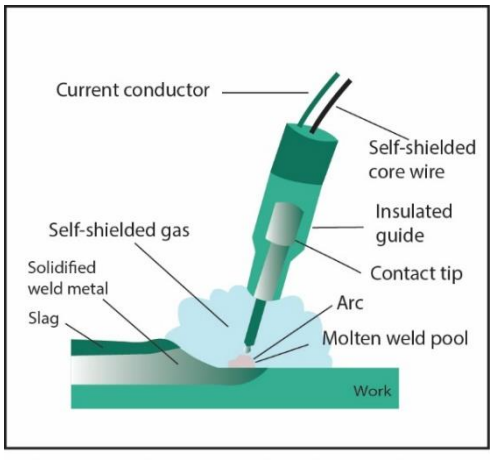


Figure 30: Flux-cored Arc Welding

FCAW uses a tubular wire with a flux core, which serves multiple purposes such as providing shielding gas,

stabilizing the arc and adding alloying elements to the weld metal. During welding, the flux decomposes, producing fumes and a gas shielding, while the high temperatures vaporize metals from both the wire and to a much lesser extent, the workpiece. These metal particles react with shielding gases to form oxides, which are a significant component of the welding fume. FCAW can use shielding gas or rely on the flux alone for protection. The fumes typically contain complex oxides, including iron oxides, manganese, nickel, and chromium, depending on the wire's composition.

All forms of the flux cored arc process and the GMAW or MIG process uses similar welding equipment (power source and wire feeder).

FCAW wire have classification which vary according to their strength and composition of the flux core. These are detailed in the following sections.

Gases typically evolved during flux cored welding include the components of any shielding gas used, O_3, CO_2/CO and some NOx.

Gas Shielded

Gas shielded flux cored welding uses a cored or tubular wire which is shielded by an active or mixed gas such as CO_2 or an argon-based gas with 10-25% CO_2 which may also contain small amounts of O_2. The tubular wire will typically contain some fluxing compounds mixed with rutile (type T1 or T2 wires), or basic oxides and fluorides ("basic" or T5 wires). Rutile type wires have excellent all-positional running characteristics and are commonly used in Queensland's coal mines.

Gas-shielded flux cored wires filled mainly with metal powders are described as "metal cored" (type T15). They are high deposition wires with running characteristics similar to those of the solid wire MIG process and are mainly used in the down hand or flat position.

Gas shielded flux cored wires produce elevated levels of welding fume, compared with gas metal arc MIG processes.

Self-Shielded

Some flux cored wires do not require shielding gas and are described as either self-shielded or "gasless". In addition to fluxing agents (e.g. oxides, borates, silicates, fluorides etc), these wires contain high levels of materials that either react with O_2 (e.g. aluminium powder) or decompose in the arc (e.g. carbonates, fluorides etc) to form a gas shield mainly consisting of CO_2/CO. Being self-shielded, they offer good weld pool protection in windy conditions. These consumables produce high levels of welding fume, when compared with gas shielded.

Manual Metal Arc Welding (MMAW)

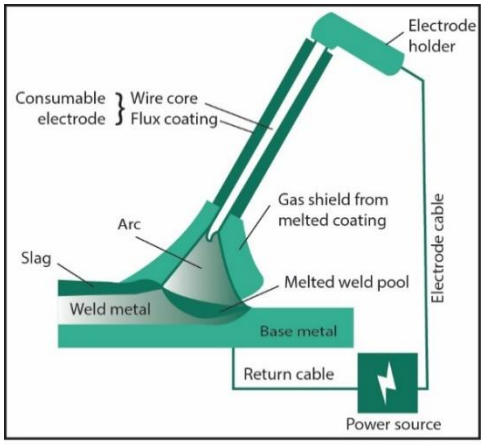


Figure 31: Manual Metal Arc Welding

MMAW also known as stick welding, uses a consumable electrode coated with metallic powders, minerals, fluxing agents and binders. When the electrode "burns" during welding, this coating either melts or decomposes, releasing gases and particulates that form a significant part of the fumes. These fumes help stabilize the arc and protect the weld area.

Coating types contain cellulose (types EXX10 or EXX11) or a range of oxides, fluorides and metal powders depending on their classification.

The high temperature of the arc melts both the electrode and the workpiece, vaporizing metals that react with O_2 to form metal oxides, which are visible as fumes. The fumes typically contain a mix of metallic oxides, such as iron and manganese, and potentially hazardous elements like chromium and nickel, depending on the electrode composition. The electrode's composition and coating significantly influence the type and quantity of fumes produced. Gases typically formed include CO_2, CO, O_3 and oxide of nitrogen. MMAW is referred to as shielded metal arc welding or SMAW in the United States of America.

Gas Tungsten Arc Welding (GTAW)

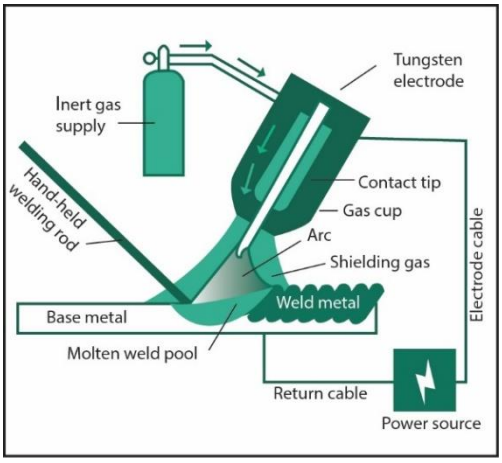


Figure 32: Gas Tungsten Arc Welding

In GTAW, also known as TIG welding, the tungsten electrode is protected by an inert gas such as argon or helium and not consumed in the weld. There are a number of variations on the process relating to the use of filler metals including the autogenous (no filler wire) and hot-wire (electric current passing through the filler wire) process, however the most common form is to add filler metal to the molten weld pool by dipping a hand-held rod into the molten pool. Due to the risk of electric shock, the boilermaker uses a technique that avoids direct contact between the filler rod and the arc.

Fumes are primarily produced from the vaporisation of the weld pool (i.e. a mix of filler wire, the metals being welded and any coatings present on them). The intense heat of the welding arc can vaporise metal surfaces, which condense into fine airborne particles, forming fumes. While the tungsten electrode typically does not contribute to fume formation because it doesn't melt. The inert shielding gas, often argon or helium, protects the weld area from atmospheric gases but does not directly create fumes. High temperatures in GTAW can lead to the formation of metal oxides, especially when welding materials like aluminium, which can produce noticeable white smoke. The use of filler materials can also contribute to fumes, with the specific metals in the filler (such as chromium, nickel, or manganese) determining the fume composition and quantity. This process typically produces very low levels of fume.

Gases evolved by the process are mainly O_3 and some NOx due to the effect of the UV radiation on O_2 in the ambient air.

Air Arc Gouging

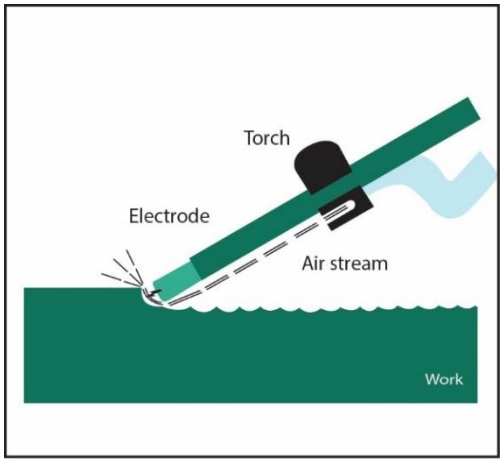
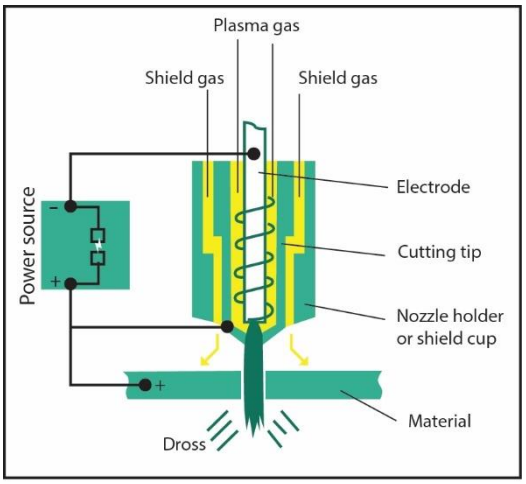


Figure 33: Air Arc Gouging

Air-arc gouging is a thermal gouging process that uses an arc from copper coated carbon electrode and a stream of compressed air to remove the molten metal and form a groove. The process runs at high amperage and produces high volumes of fumes. Fume composition is similar to the material being gouged with some traces of copper. Some gases are also created which consists of CO_2/CO, O_3 and NOx.

Plasma Cutting and Gouging



Plasma cutting and gouging is a thermal process that uses a constricted plasma arc where a high velocity jet of similar in composition to the material being cut or gouged with gases evolved being mainly O_3 and some NOx.

Plasma cutting can occur on a cutting table or in some cases over a water table (or underwater) which captures most of the welding fume. Plasma cutting underwater is essentially fume-free.

Figure 34: Plasma cutting and gouging

Thermal Cutting and Gouging

Thermal cutting and gouging uses an O_2/fuel gas (e.g. acetylene or Liquefied Petroleum Gas) mix to heat the metal being cut to ignition temperature. A jet of O_2 causes the metal to oxidise and the molten swarf is removed by the cutting jet. Particulates in the fume are oxides similar in composition to the material being cut or gouged. Gases emitted include the products of combustion and remnant O_2 that is not consumed in the cutting or gouging process,

Brazing and Soldering

Brazing when used in Queensland coal mines can use either an O_2/fuel gas flame or an electric arc process similar to GMAW or *MIG* (i.e. MIG Brazing). In all cases, unlike welding, the material being joined does not melt. Filler material that has a melting temperature >450°C is described as a brazing material and if <450°C, it is considered a soldering process. Silver soldering is a process used within mines; however, it is a brazing process by definition. Fumes and gases emitted during brazing are minimal other than in arc brazing. Most braze alloys are copper based with silver solders for example having a range of copper/zinc/silver alloy compositions. Liquidus temperature for silver solders ranges from 655°C to 760°C depending on composition.

Appendix C: Sample record sheet example

First Name:	Last Name:	Location:		
Date:	Shift:	Crew:	Hrs per Shift:	Roster:
	Day / Afternoon / Night	A B C D Other:		
SEG: e.g. HME Refurbishment		Job Title: e.g. Boilermaker	Contractor:	Yes / No
			Confined Space:	Yes / No
Sample Type:	Full Shift / Task Based	Total Arc On Time Over Full Shift:		

Personal Protective Equipment (PPE)			
Respirator Worn:	Disposable half- face/ Reusable half-face/ Reusable full-face/ PAPR/ Airline/ None/ Other:	Respirator Model:	
		Filter Type/ Class	Particulate: P2 P3 Gas: A1 B1 E1 Odour
Work Conditions:	Normal / Not Normal Comment:		
Ventilation Controls Used:	Dilution (Pedestal/mechanical fan) Local Exhaust Ventilation (LEV) On Torch Extraction Natural		
Brand/ Model/ Type of Ventilation/ Extraction system (if used)			

PLEASE LIST ***ALL*** ACTIVITIES PERFORMED THROUGHOUT YOUR SHIFT/SAMPLING PERIOD INCLUDING RELEVANT TIMES

Start-Finish Time	Task	Vent Controls Used	Arc-On Time	Welded Material	Welding Process	Consumable	Sampler on only during arc-on time
e.g. 0600-0700	*E,g, Repair of handrail*	*E.g. On Torch*	*E.g. 15 minutes*	*E.g. Mild Steel*	*E.g. Flux Core*	*E.g. Verticore 3XP*	Yes/ No
							Yes/ No
							Yes/ No
							Yes/ No
							Yes/ No
							Yes/ No
							Yes/ No
							Yes/ No
							Yes/ No
							Yes/ No
							Yes/ No
							Yes/ No
							Yes/ No

At any time during the sampling period, did the sampler:	If yes, please comment below:
Get turned off when not welding? (required to exclude other contaminants)	Yes/ No
Get turned off when grinding? (required to exclude grinding dust)	Yes/ No
Be exposed to other contaminants besides welding fume? E.g. coal dust, grinding dust, wood dust, metal shavings, spray paint etc.	Yes/ No
Get dropped, damaged or severely impacted?	Yes/ No
Become significantly wet, exposed to or immersed in water?	Yes/ No
Have pump, tubing or sample head disconnected?	Yes/ No
Sampling head inlet hole come into direct contact with dusty/ dirty surface?	Yes/ No
Was the sampling head removed from the breathing zone at any time?	Yes/ No
Get removed at any time?	Yes/ No
Did anything else occur which could have compromised the sampling process?	Yes/ No

Please comment on the following exposure controls:	If absent or faulty, please comment below:
Ventilation controls in use? E.g. LEV, On-torch extraction	Adequate / Absent / Faulty / NA
Enclosures/ covers/ barriers in place?	Adequate / Absent / Faulty / NA
Ventilation fume extraction hood in use, positioned close to weld?	Adequate / Absent / Faulty / NA
Were any other normally present exposure controls absent or faulty?	Yes/ No
Did any unusual events take place which may contribute to a higher or lower exposure than normal? E.g. unusual activities, confined space, multiple welders etc	Yes/ No
Weather conditions during assessment period (e.g. still air, light breeze, somewhat windy, very windy, raining etc)	

Diagram of work area showing job, ventilation and airflow direction AND/OR additional comments e.g. controls in operation, barriers etc.

Please sign here as acknowledgement that you understand that when full shift monitoring, sampling equipment is not to be fitted or removed/disconnected without a hygiene technician present, and that the information provided on this sheet is an accurate reflection of the conditions present during the sampling period to the best of your knowledge.

Name: _____ Signature: _____ Date: _____

Minimum information requirements to be recorded by CMW

Worker information	Shift information
Full name Employment status (if contracting, include company) Job title SEG Roster details including shift length and crew.	Tasks undertaken, including timeframe (start and end), duration and location for whole of shift. Equipment used Welding method used Welding consumable(s) used (incl. brand and model name/number) Base metal being welded Use and type of personal protective equipment Welding fume controls available and used. Operating as intended (Y/N). Whether the shift was representative of normal work activities performed. Any events or conditions that may have contributed to significant fume exposure

Minimum information requirements to be recorded by competent person

Sample information	Environmental and operational information
Name of competent person undertaking the sample. Laboratory identifier, filter number and size selective sampler ID (if applicable). Sample date, sample time on and off, total runtime, pre, post and average flow rate. Equipment used (i.e. pumps and calibrator asset/serial number). Size selective sampler type used (ie inhalable IOM, respirable cyclone). Position of sampler (ie. Inside/ behind weld shield, outside/ behind weld shield on shirt collar) Sample validity.	Controls in use and their effectiveness. Location of task (indoors, outdoors, confined space). Ventilation type used. Wind conditions. Housekeeping details. Exposure sources. Operational conditions. Weather conditions or seasonal factors that could have impacted sample.

Appendix D: Welding fume contaminants

Welding Process	Common Welding Tasks	Common Consumables	Material	Typical Fume Components
MMAW	Bucket repairs, dump truck tray repairs, dragline tub, CHPP wear plates	Ferrocraft 61 Conac 49C Easyarc 7016 Austarc 16TC	Mild steel	Fe, Mn, Cr, Ni, Cu
	Dragline house, handrail repairs, access ladder/walkway repairs	Satincraft 13 Omnia 46 GP		
			Stainless steel	Fe, Mn, Cr, Ni, Cr(VI), Mo, O_3
	Hardfacing	Cobalarc 350 Wearshield BU 30	Cr-containing consumables	Fe, Mn, Cr, Ni, Cr(VI), Co, O_3
			Cr-free consumables	Fe, Mn, Ni, Co, B, O_3
			Austenitic Mn High carbon austenitic steels	Fe, Mn, O_3 Fe, Mn, Cr, Cr(VI), O_3
GMAW (with solid wire)	Dragline house, handrail repairs, access ladder/walkway repairs	Autocraft LW1 Ultramag S6	Mild steel	Fe, Mn, Cr, Ni, Cu
			Stainless steel	Fe, Mn, Cr, Ni, Cr(VI), O_3
FCAW-GS (with metal-cored and flux-cored wire – gas shielded)	Bucket repairs, dump truck tray repairs, dragline tub, excavator boom repairs, excavator dipper arm repairs, CHPP wear plates, CHPP feeder repairs	Verti-Cor 3XP Primacore-MW71 Verti-Cor 81Ni 1 Primcore-81NiH	Mild steel	Fe, Mn, Cr, Ni, Cu
	Dragline bucket trunnions, CHPP wear plates, CHPP feeder repairs	Verti-Cor 309LT CLEAROSTA F 309L Verti-Cor 316LT Tetra V 316L-G	Stainless steel	Fe, Mn, Cr, Ni, Cr(VI), O_3
	Hardfacing		Cr-containing consumables	Fe, Mn, Cr, Ni, Cr(VI), Co, O_3
			Cr-free consumables	Fe, Mn, Ni, Co, B, O_3
			Austenitic Mn High carbon austenitic steels	Fe, Mn, O_3 Fe, Mn, Cr, Cr(VI), O_3
FCAW-SS (with flux-cored wire – self shielded)			Mild steel	Fe, Mn, Cr, Ni, Cu, Al
			Stainless steel	Fe, Mn, Cr, Ni, Cr(VI), Al, O_3
	Hardfacing		Cr-containing consumables	Fe, Mn, Cr, Ni, Cr(VI), Co, Al, O_3

Welding Process	Common Welding Tasks	Common Consumables	Material	Typical Fume Components
			Cr-free consumables	Fe, Mn, Ni, Co, B, Al, O_3
			Austenitic Mn High carbon austenitic steels	Fe, Mn, Al, O_3 Fe, Mn, Cr, Cr(VI), Al, O_3
GTAW (autogenous or with filler material)			Mild steel	Fe, Mn
			Stainless steel	Fe, Mn, Cr, Ni, O_3
Arc/ Plasma Gouging			Mild steel	Fe, Mn, O_3
			Stainless steel	Fe, Mn, Cr, Ni, Cr(VI), O_3

Appendix E: Common welding tasks and fume control options

Fume Control Options	1. Substitute process	2. Ventilation	3. Respiratory protection
	1A. Weld method MIG	2A. On torch LEV	3A. Disposable - P2
	1B. Waveform control	2B. Fume hood LEV	3B. Half face - P2
	1C. Shield gas < 18% C02	2C. General - mechanical	3C. PAPR - P2
	1D. Electrode (Cu Free, Cr Free)	2D. General with Ventilation plan	3D. Air line/air supply
	1E. Plasma gouging		
	1F. Robot cobot		

NOTE: A combination of different control options will typically need to be applied in practice, in order to manage welding fume exposure effectively.

Common welding tasks in Queensland coal mining	Current weld method(s)			Fume control options			
	Consumable Type	AS/NZS ISO classification	AWS classification	Method	Substitute process	Ventilation	Respiratory protection
Dump truck tray	Rutile FCAW	B T 493 T12 1 M A N2 U H10	E71T-1/12M H8	FCAW	1A, 1A + 1B, 1C	2A, 2B, 2C	3B, 3C
	Rutile FCAW	B T 55 4 T1 1 M A N2 U H10	E81T1-1 Ni1 H8	FCAW	1A, 1A + 1B, 1C	2A, 2B, 2C	3B, 3C
	Low hydrogen stick	B E4918 A U H10	E7018	MMAW		2b, 2C	3B, 3C

Common welding tasks in Queensland coal mining	Current weld method(s)				Fume control options		
Bucket repairs	Rutile FCAW	B T 493 T12 1 M A N2 U H10	E71T-1/12M H8	FCAW	1A, 1A + 1B, 1C	2A, 2B, 2C	3B, 3C
	Rutile FCAW	B T 55 4 T1 1 M A N2 U H10	E81T1-1 Ni1 H8	FCAW	1A, 1A + 1B, 1C	2A, 2B, 2C	3B, 3C
	Low hydrogen stick	B E4918 A U H10	E7018	MMAW		2B, 2C	3B, 3C
Dragline bucket trunnions	Stainless FCAW	TS309L FM1	E309LT1-1/4	FCAW		2A, 2B, 2C	3B, 3C, 3D
Dragline car body house (mild)	Rutile low strength stick	B E4313 A	E6013	MMAW		2B, 2C	3B, 3C
	Solid wire	B G 49A 3U C1/M21/M24 S4	ER70S-4	GMAW	1B, 1C	2A, 2B, 2C	3A, 3B
Dragline tub and revolving frame	Rutile FCAW	B T 493 T12 1 M A N2 U H10	E71T-1/12M H8	FCAW	1A, 1A + 1B, 1C	2A, 2B, 2C, 2D	3C, 3D
	Rutile FCAW	B T 55 4 T1 1 M A N2 U H10	E81T1-1 Ni1 H8	FCAW	1A, 1A + 1B, 1C	2A, 2B, 2C	3B, 3C
	Low hydrogen stick	B E4918 A U H10	E7018	MMAW		2B, 2C	3B, 3C
Dragline Boom repairs	MnMo GMAW	B G 55A 3 C1/M21/M24 S4M31	ER80S-D2	GMAW	1B, 1C	2A, 2B, 2C	3A, 3B, 3C
Excavator boom repairs	Rutile FCAW	B T 493 T12 1 M A N2 U H10	E71T-1/12M H8	FCAW	1A, 1A + 1B, 1C	2A, 2B, 2C	3B, 3C

Common welding tasks in Queensland coal mining	Current weld method(s)			Fume control options			
	Rutile FCAW	B T 55 4 T1 1 M A N2 U H10	E81T1-1 Ni1 H8	FCAW	1A, 1A + 1B, 1C	2A, 2B, 2C	3B, 3C
Excavator dipper arm repairs	Rutile FCAW	B T 493 T12 1 M A N2 U H10	E71T-1/12M H8	FCAW	1A, 1A + 1B, 1C	2A, 2B, 2C	3B, 3C
	Rutile FCAW	B T 55 4 T1 1 M A N2 U H10	E81T1-1 Ni1 H8	FCAW	1A, 1A + 1B, 1C	2A, 2B, 2C	3B, 3C
CHPP wear plates	Rutile FCAW	B T 493 T12 1 M A N2 U H10	E71T-1/12M H8	FCAW	1A, 1A + 1B, 1C	2A, 2B, 2C	3B, 3C
	Rutile FCAW	B T 55 4 T1 1 M A N2 U H10	E81T1-1 Ni1 H8	FCAW	1A, 1A + 1B, 1C	2A, 2B, 2C	3B, 3C
	Stainless FCAW	TS309L FM1	E309LT1-1/4	FCAW		2A, 2B, 2C	3B, 3C, 3D
	Low hydrogen stick	B E4918 A U H10	E7018	MMAW		2B, 2C	3B, 3C

Common welding tasks in QLD Coal mining	Current weld method(s)			Fume control options			
	Consumable Type	**AS/NZS ISO classification**	**AWS classification**	**Method**	**Substitute process**	**Ventilation**	**Respiratory protection**
CHPP feeder repairs	Stainless FCAW	TS309L FM1	E309LT1-1/4	FCAW		2A, 2B, 2C	3B, 3C, 3D
	Stainless FCAW	TS316L FM1	E316LT1-1/4	FCAW		2A, 2B, 2C	3B, 3C, 3D
	Rutile FCAW	B T 493 T12 1 M A N2 U H10	E71T-1/12M H8	FCAW	1A, 1A + 1B, 1C	2A, 2B, 2C	3B, 3C
	Rutile FCAW	B T 55 4 T1 1 M A N2 U H10	E81T1-1 Ni1 H8	FCAW	1A, 1A + 1B, 1C	2A, 2B, 2C	3B, 3C
Guarding, Handrail Repairs (mild)	Rutile low strength stick	B E4313 A	E6013	MMAW		2A, 2B, 2C	3B, 3C
	Solid wire	B G 49A 3U C1/M21/M24 S4	ER70S-4	GMAW	1B, 1C	2A, 2B, 2C	3A, 3B
	16TC			MMAW			3B, 3C
Access ladder/walkway repairs (mild)	Rutile low strength stick	B E4313 A	E6013	MMAW		2A, 2B, 2C	3B, 3C
	Solid wire	B G 49A 3U C1/M21/M24 S4	ER70S-4	GMAW	1B, 1C	2A, 2B, 2C	3A, 3B
Aluminium repairs		AL5356	AL5356	GTAW		2B, 2C	3B, 3C
		AL5356	AL5356	GMAW		2A, 2B, 2C	3B, 3C
Hardfacing	Hardfacing	2576: 1435-A4	-	MMAW	1D	2B, 2C	3B, 3C, 3D
Arc Air Gouging	Carbon Rods			AAG	1E	2C	3B, 3C, 3D
Welding inside confined space					1F	2D, 2D + 2A, 2D+2B	3C, 3D

Appendix F: PAPR checklist (example)



PAPR Maintenance Inspection Checklist

Date:

Date cartridge /filter expires:

Person conducting check:

If PAPR is found to be faulty remove from service and arrange for repair.

VISUAL INSPECTION

☐	Check for visible damage. Ensure all parts, including the hood, helmet, facepiece and hoses are in good condition. (remove sheath on hose and stretch hose to check for cracks and holes)	☐	Verify the hose connections are secure. Verify all parts are from the same manufacturer.

BATTERY CHECK

☐	Ensure the battery is fully charged. Inspect battery for any damage.	☐	Confirm battery is properly installed and secure.

FILTER / CARTRIDGE CHECK

☐	Verify that the correct particulate filter or gas cartridge is installed for the specific contaminants. Inspect the filter or cartridge for any damage or past expiry date.	☐	Replace the filter or cartridge if it is damaged or past expiry date. For gas cartridges, dispose 6 months after opening (regardless of use or expiry date) New filters should be in clean, sealed, unopened package.

AIRFLOW CHECK

☐	Ensure the appropriate filter/ cartridge is installed and remove any filter/ cartridge caps. Turn on the PAPR and ensure that it is operating correctly.	☐	Use a flow meter to measure the airflow. Detach hose from helmet. Attach air flow indicator tube to helmet end. Turn on belt pack to lowest setting If the ball does not rise above the line continue fault finding.

ALARM CHECK

☐	Cover the air outlet of the hose with your hand until an audible or visual warning is triggered. Remove your hand and alarm should stop.	☐	Check any alarms or indicators for proper operation.

HEADPIECE CHECK

☐	Inspect the headpiece (helmet or hood) for any damage. Ensure the face seal or shroud is intact and not damaged. Verify that the headpiece fits properly and forms a good seal.		Replace any worn components including face seals, headbands etc.

Appendix G: LEV checklist (example)



LEV Maintenance Inspection Checklist

Date:

Person conducting:

Ensure LEV is powered down prior to starting check.

VISUAL INSPECTION

☐	Check for visible damage. Ensure all parts, including the ducting, hood, extraction arms, connections and seals are in good condition. Check ducting for splits, burn holes or other damage. Stretch flexible ducting to check between pleats for holes / cracks.	☐	Inspect fans, fan covers/grilles, electrical cables and switches/buttons for signs of damage or wear.
☐	Inspect interior of ducting and hoods to ensure clean and free of excessive dust build-up. Check low spots/elbows for accumulated dust/blockage.	☐	Ensure any dampers present are operational and adjusted to maintain proper airflow through system.
☐	Inspect belts, bearings, pulleys (where fitted) for splits and other wear and tear	☐	Inspect guarding for damage

AIR CLEANER/FILTER CHECK

☐	Verify that the correct air cleaner/particulate filter is installed. Inspect the cleaner for any damage. Check OEM filter expiry date.	☐	Replace the cleaner if damaged or expired.
☐	Check seals around air cleaner housing are present and intact.	☐	Check accumulation of dust on air cleaner or in dust collector. Significant dust accumulation to be removed while using PAPR. Clean reusable air filters as per OEM instructions. Disposable filters to be disposed of and replaced.

FUNCTION CHECK

☐	Ensure the appropriate filter is installed. Turn on the LEV and ensure that it is operating correctly with no unusual noise or vibrations. Check any alarms or indicators for proper operation	☐	Confirm airflow filters are operational and showing system performance (where fitted). Manually operate filter self-clean function (where fitted) to ensure correct operation.

AIRFLOW CHECK- FOR ON TORCH TYPE LEV

☐	Check airflow at nozzle using airflow meter (supplied by OEM or equivalent) to ensure flow rate meets OEM specs.

AIRFLOW CHECK- FUME HOOD TYPE LEV

☐	Check for any alarms or indicators for proper operation. Competent person to check airflow / velocity at hood face or within ducting using an anemometer to verify airflow volume meets OEM specifications.

THIS CHECKLIST IS INTENDED AS A GENERAL GUIDE– IT SHOULD BE MADE SPECIFIC TO THE TYPES OF LEV USED ON SITE.

Appendix H: Confined space: Example ventilation plan drawings

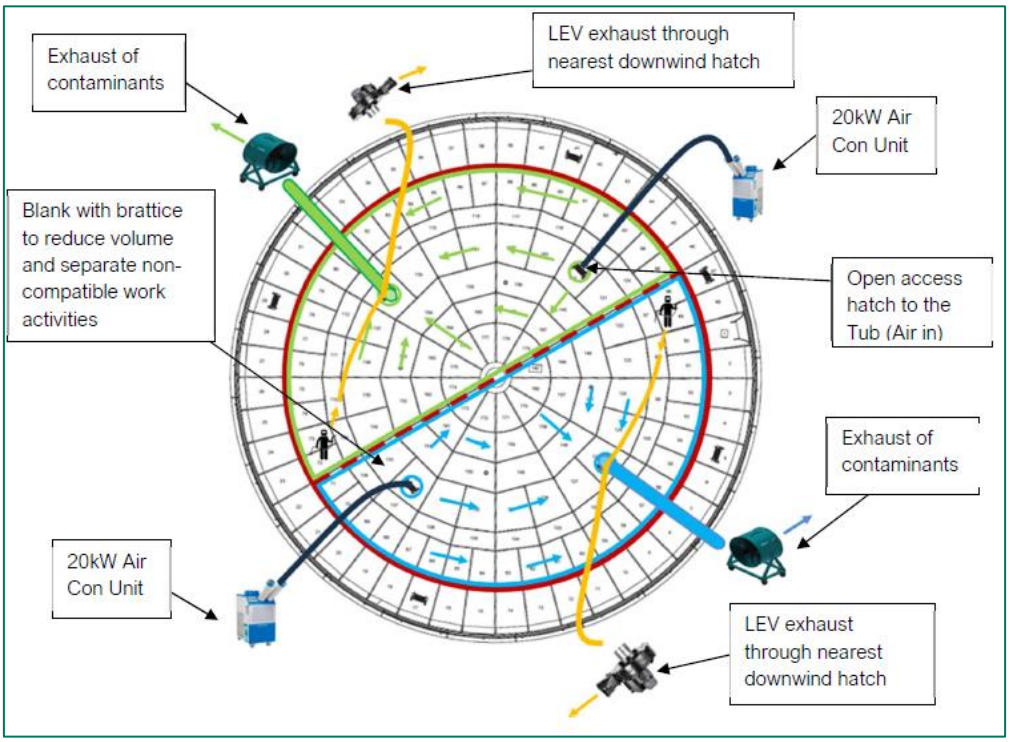


Figure 35: Dragline tub outer ring ventilation plan (Field Engineers 2025)

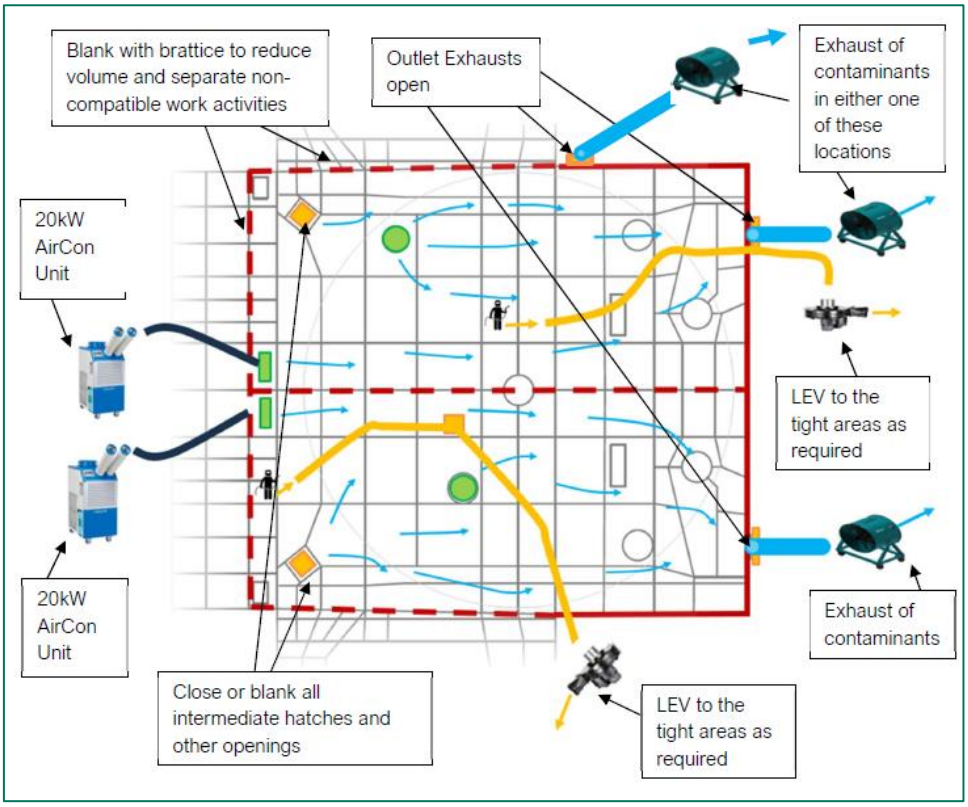


Figure 36: Dragline tub revolving frame forward compartment ventilation plan (Field Engineers 2025)

Important note: Performance of the ventilation set up should be tested and verified after installation and prior to use.

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Appendix J: Additional resources

SafeWork Australia

Code of practice: *Welding processes*

Standards Australia

AS 1674.1 *Safety in Welding and allied processes Part 1: Fire precautions*

As 1674.2 *Safety in Welding and allied processes Part 2: Electrical*

Weld Australia

Technical Note 7: Health and safety in welding

Technical Note 22: Welding electrical safety

Technical Guidance Note TGN-SW01: Fume minimisation guidelines

Technical Guidance Note TGN-SW03: Welding electrical safety – Electric shock injury and electrocution reporting

Technical Guidance Note TGN-SW07: Health and safety in welding – Guides and forms